



中國人民大學
RENMIN UNIVERSITY OF CHINA

社会网络分析基础

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基础概念

社会网络类型

节点测度指标

路径搜索

社群发现

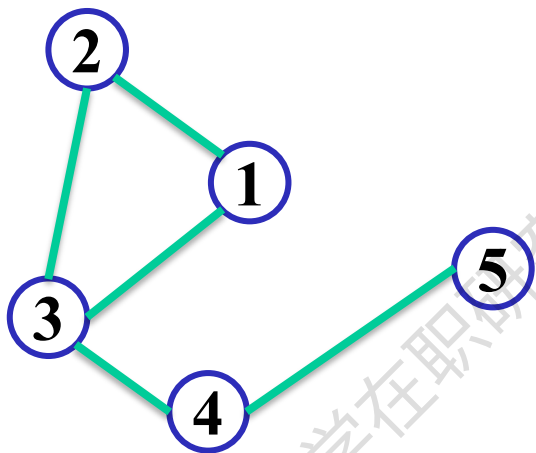
Gephi 实操

图论 Graph theory

略枯燥，尽量举例说明



图形表示, 矩阵表示, 三元组表示



Node	Feature 1	Feature 2	Feature 3	Feature 4	Feature 5
1	1	1	1	0	0
2	1	1	1	0	0
3	1	1	1	1	0
4	0	0	1	1	1
5	0	0	0	1	1

(1, 2, 1)

(1, 3, 1)

(2, 3, 1)

(3, 4, 1)

(4, 5, 1)

✓ 依据**研究群体**分类：自我中心网络（Ego-centered Networks）VS整体网络（Whole Networks）

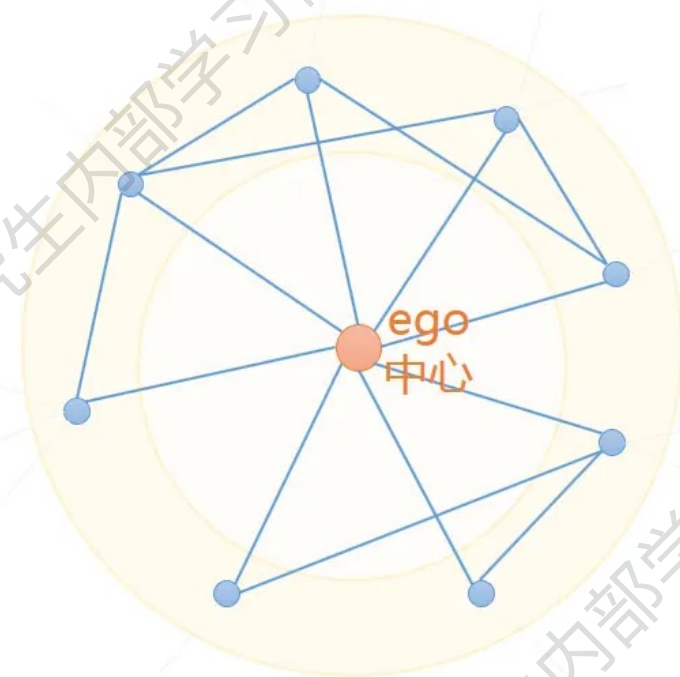
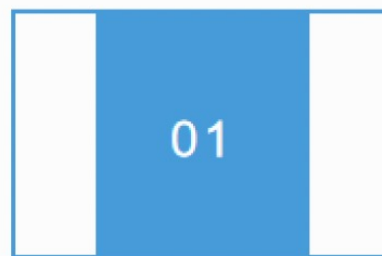
01

自我中心网络是从**个体**的角度来界定社会网络，以特定行动者为研究中心，主要考虑与该行动者相关的联系，以此来研究个体行为如何受到其人际网络关系的影响。则分析个体成员之间关系的异质性、同质性、结构洞等。

02

而整体网络关注的焦点则是网络整体中角色关系的综合结构或群体中不同角色的关系结构。一般关注密度、互惠性、关系的传递性、子群结构、核心-边缘结构等。

✓ 依据**研究群体**分类：自我中心网络（Ego-centered Networks）VS整体网络（Whole Networks）



研究中心，主要考
网络关系的影响。

体中不同角色的关
系-边缘结构等。

图1：Ego Network示例。橙色节点为中心节点，是ego network的中心，周边的蓝色节点都与此节点直接相连；图中所有节点与他们之间的相互连边（蓝色）便构成了橙色节点的ego network。

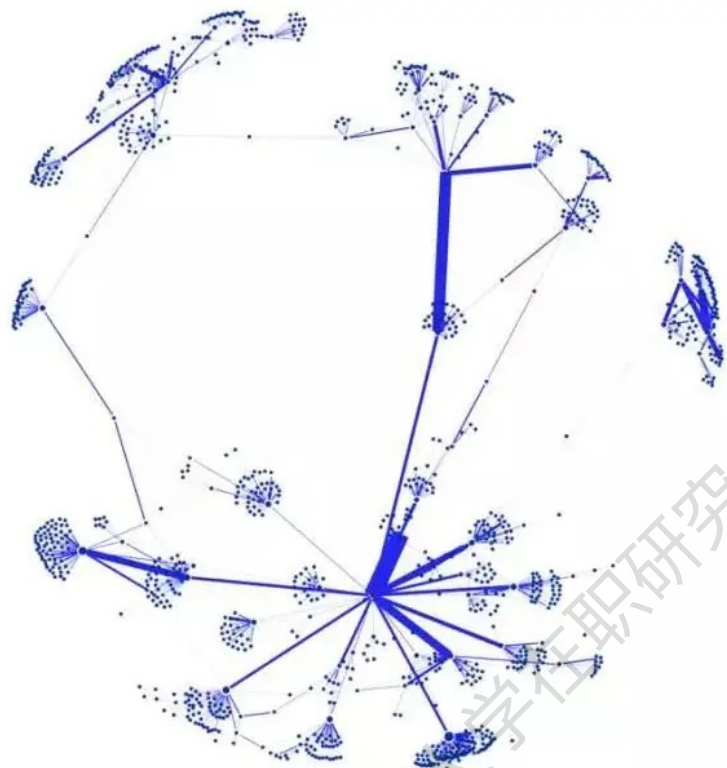
- 通过分析用户电信网络通话记录，处理成无向图，用户可以分为普通用户、某大型企业员工、传销组织人员等

提取不同类别用户的ego network，识别传销组织人员

Ego network应用案例-非法传销网络



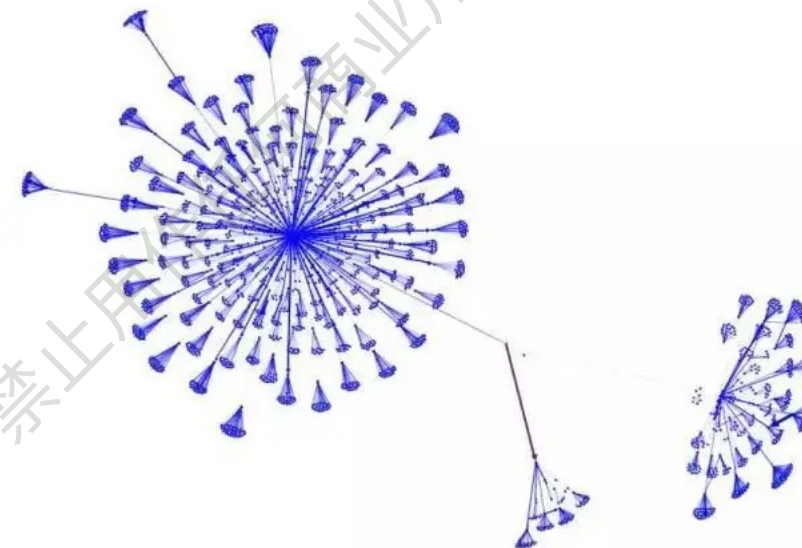
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普通用户



某企业领导的ego network
(局限在本企业员工)



某非法传销组织头目的ego network

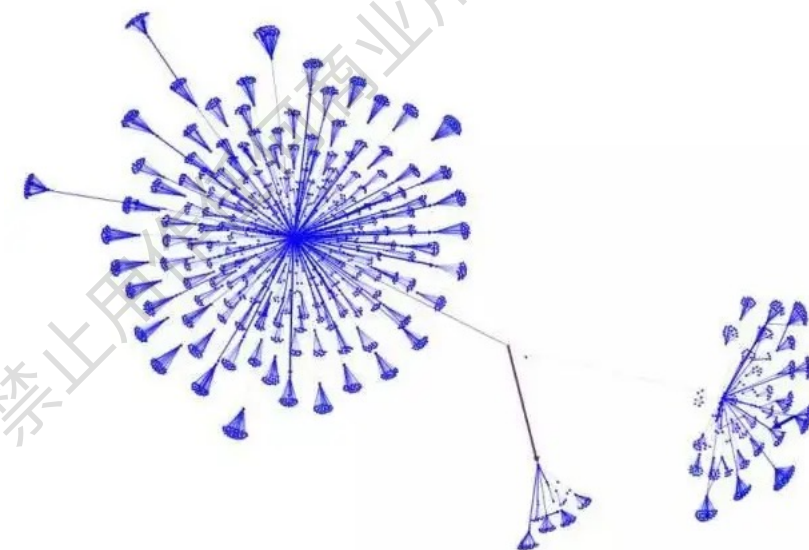
Ego network应用案例-非法传销网络



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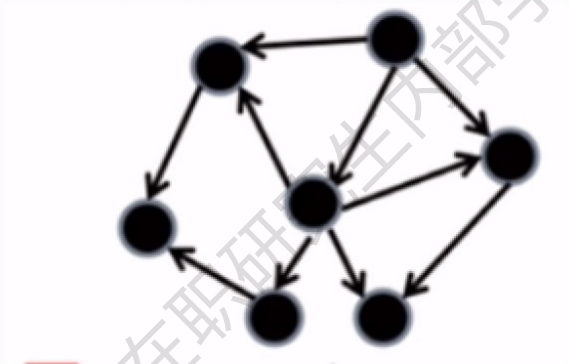
特点:

- 节点之间不跨级联系;
- 同级不抢占下线;
- 同级非同上下线节点不联系

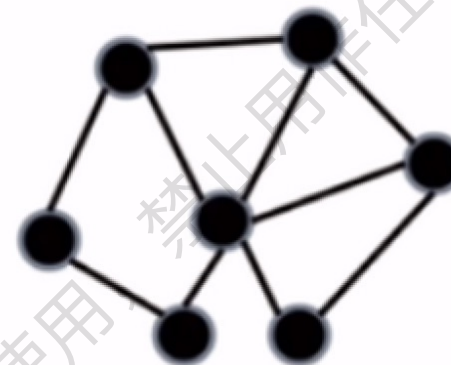


某非法传销组织头目
的ego network

✓ 依据**关系方向**分类：有向图（Directed） VS 无向图（Undirected）（对称关系）



EG：关注关系、投资关系、引用关系...

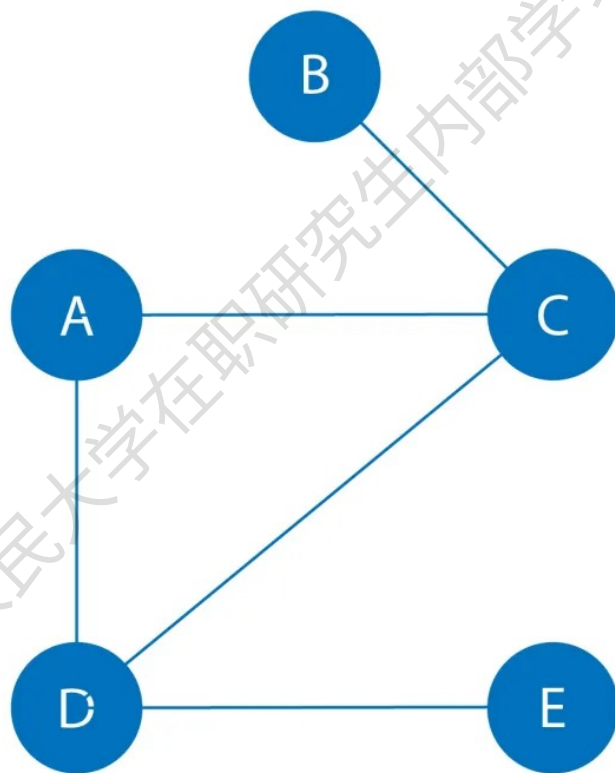


EG：好友关系、合作关系、共现关系.....

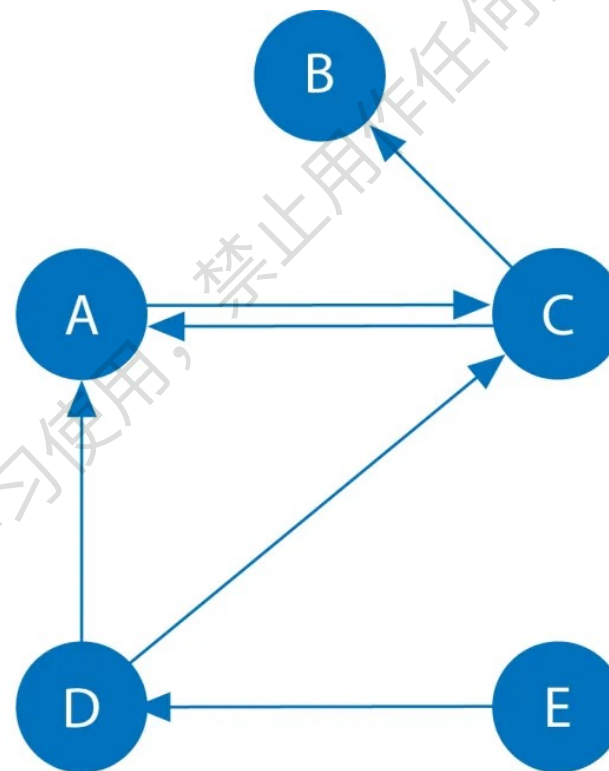
有向图示例-好友 (like) 网络



- 有向图绘制了一个简单的同学网络，边的方向代表着“喜欢”。那么从图中，我们可以知道，同学中“最受欢迎的”的人是“A”和“C”。

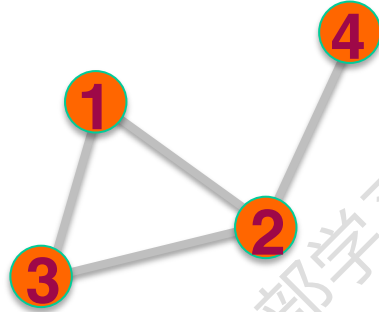


Undirected



Directed

Undirected



$$A_{ij} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

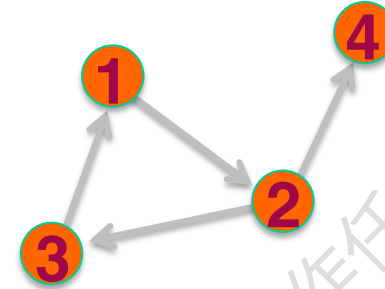
$$A_{ii} = 0$$

$$A_{ij} = A_{ji}$$

$$L = \frac{1}{2} \sum_{i,j=1}^N A_{ij} \quad \langle k \rangle = \frac{2L}{N}$$

Actor network, protein-protein interactions

Directed



$$A_{ij} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$A_{ii} = 0$$

$$A_{ij} \neq A_{ji}$$

$$L = \sum_{i,j=1}^N A_{ij} \quad \langle k \rangle = \frac{L}{N}$$

WWW, citation networks

有向图示例-专利引用网络



- P1引用P3

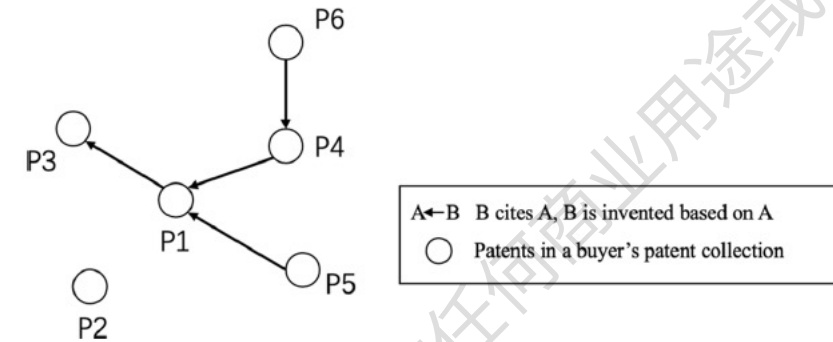


Fig. 2 An illustration of an intra-collection citation network

公司内部专利引用网络

Scientometrics
<https://doi.org/10.1007/s11192-021-04180-x>



A personalized recommendation system for high-quality patent trading by leveraging hybrid patent analysis

Wei Du¹ · Yibo Wang¹ · Wei Xu¹ · Jian Ma²

Received: 1 November 2020 / Accepted: 7 October 2021
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1-mode, 2-mode, multi-mode

✓ 依据**节点类型**分类：1-模网 VS 2-模网

• 1 模网

由一个行动者集合内部各个行动者之间的关系构成的网络。

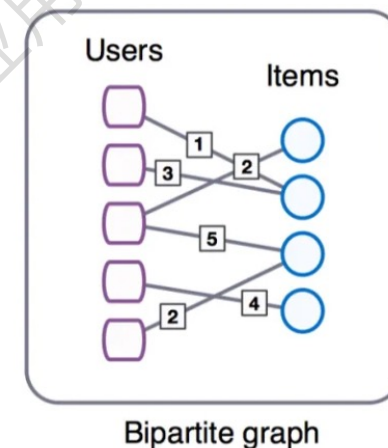
	A	B	C
A			
B			
C			



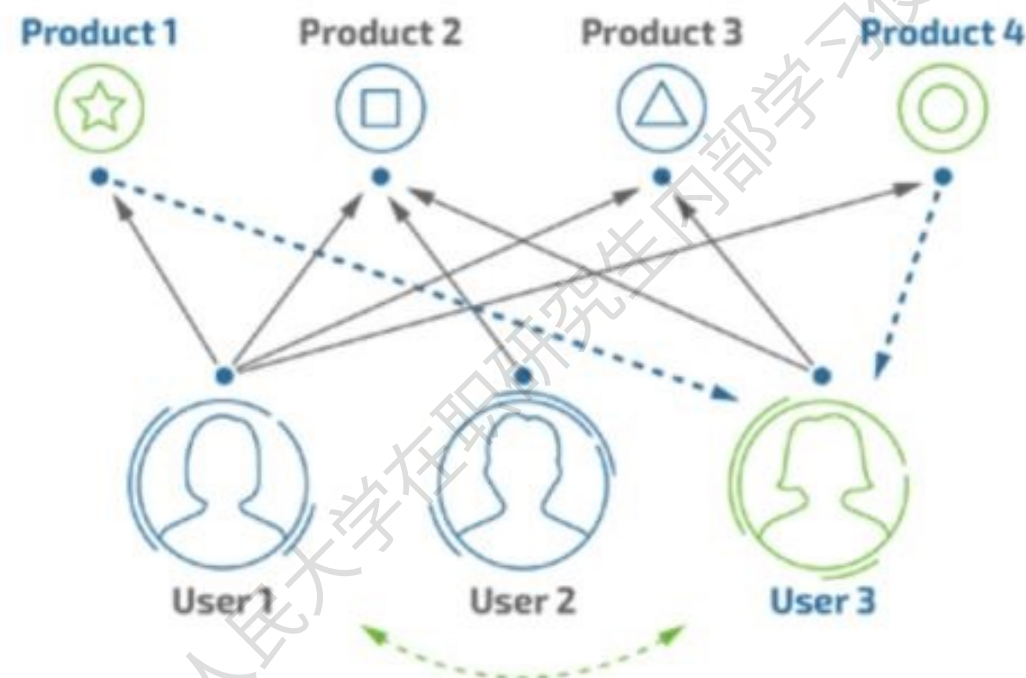
	C	D	E
A			
B			
C			

• 2-模网

一类行动者集合与另一类行动者集合之间关系构成的网络。
(eg: 隶属网络)



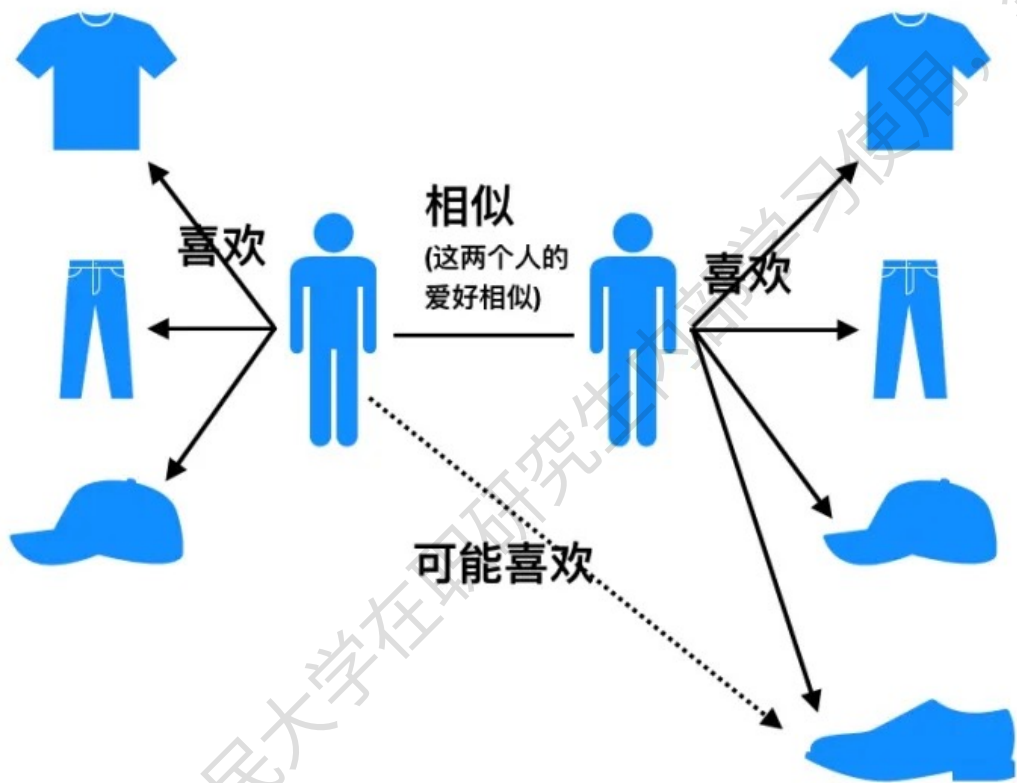
典型二分图-user-item interaction



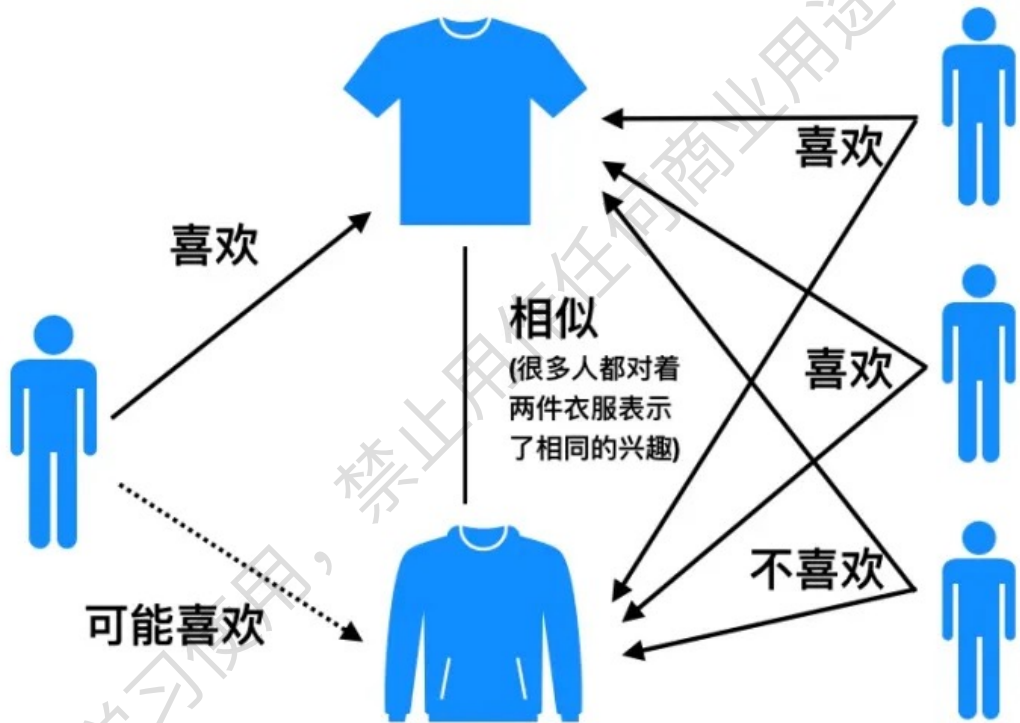
						
	1	1			1	
	?	1	?	?	1	?
			1		1	1
						
			1	1		1

collaborative: *similar interactions*

”物以类聚，人以群分“的朴素协同过滤推荐



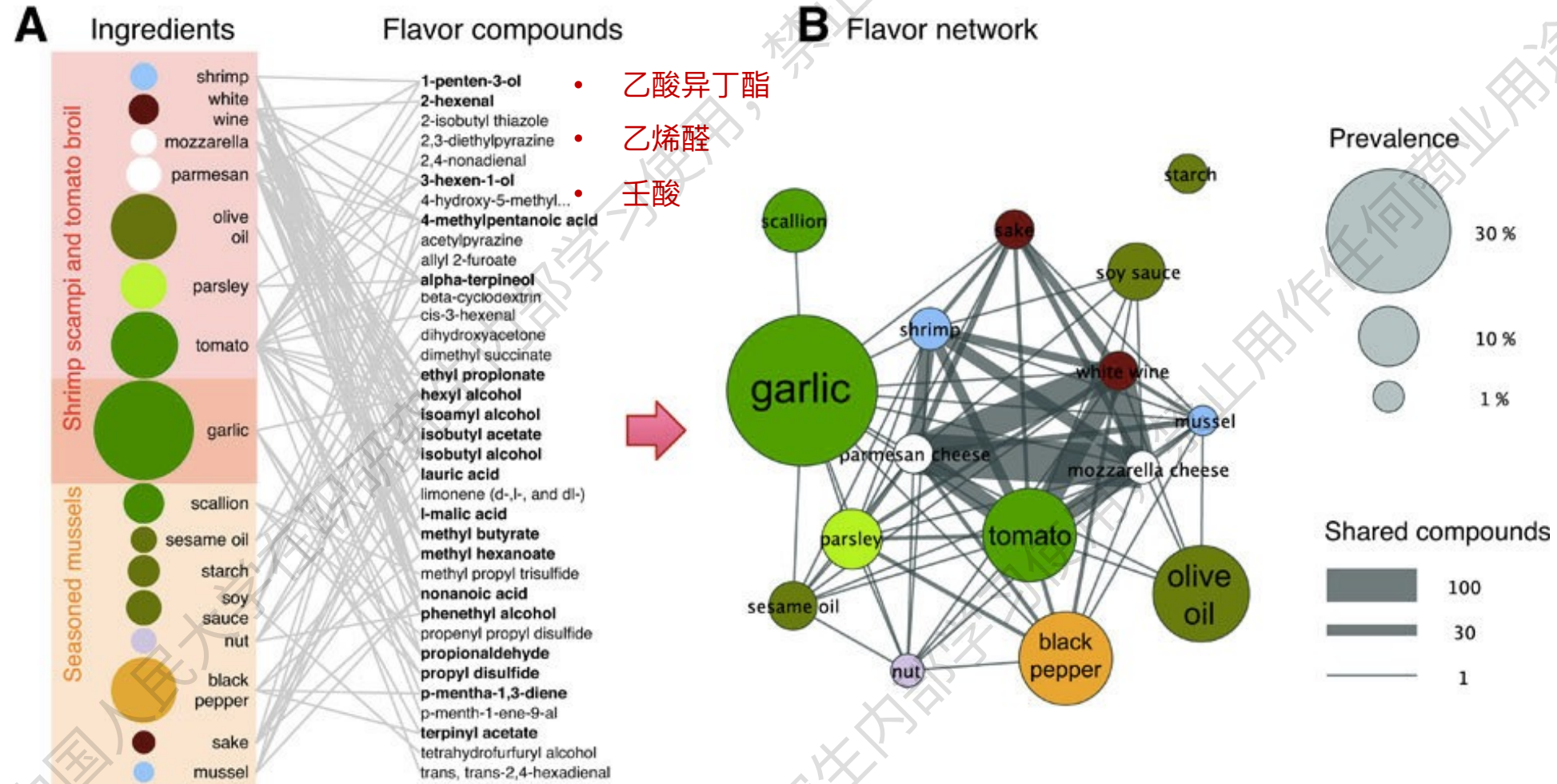
“人以群分”的基于用户的协同过滤



“物以类聚”的基于物品的协同过滤

知乎 @刘强

Ingredient-Flavor二部图

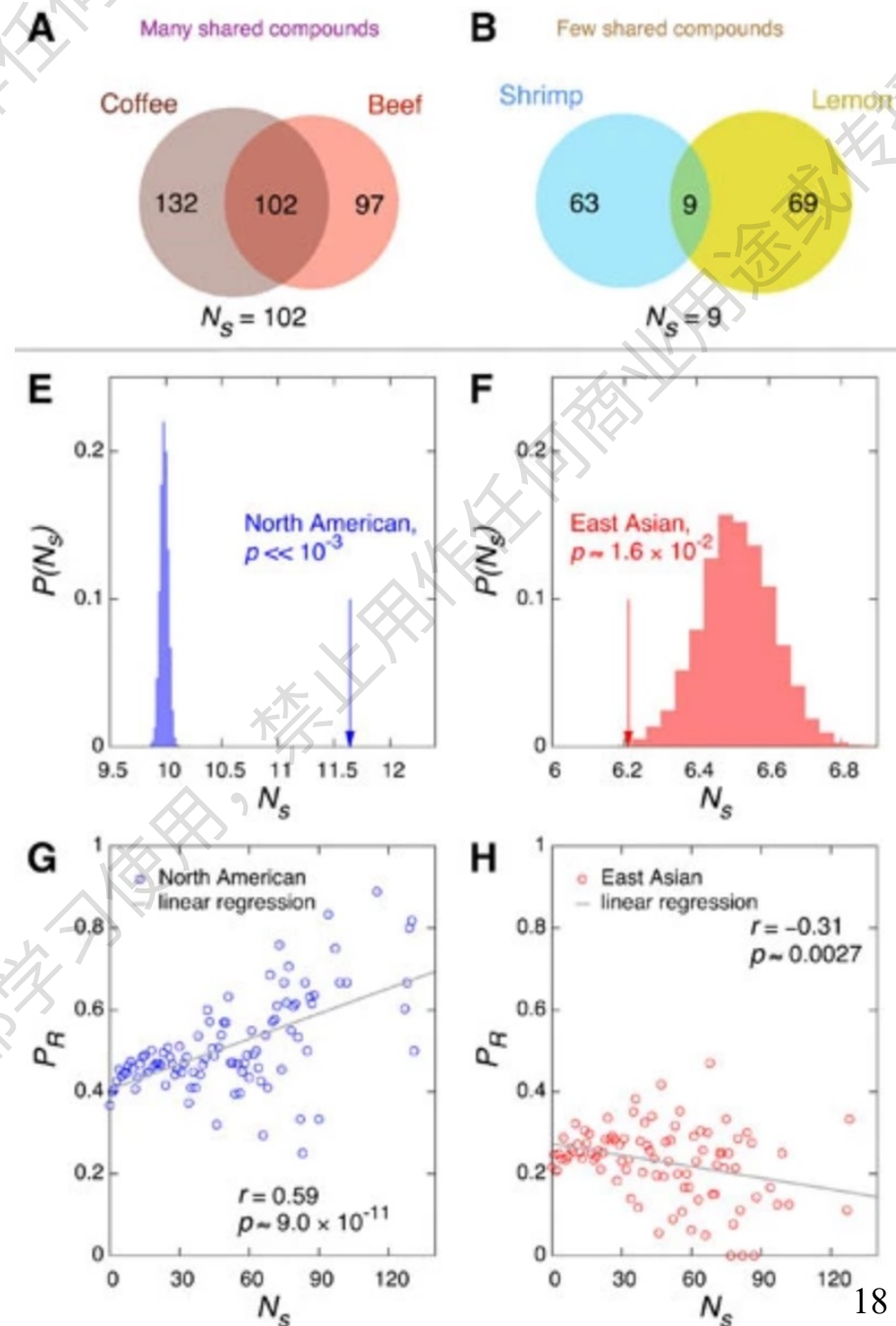


Y.-Y. Ahn, S. E. Ahnert, J. P. Bagrow, A.-L. Barabási Flavor network and the principles of food pairing , *Scientific Reports* 196, (2011).

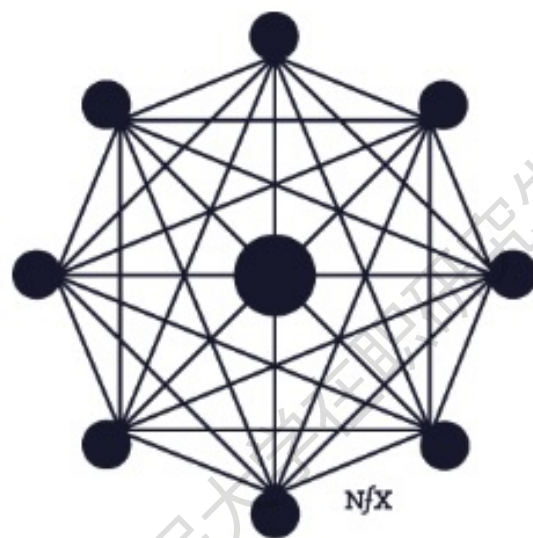
Flavor network应用

Findings

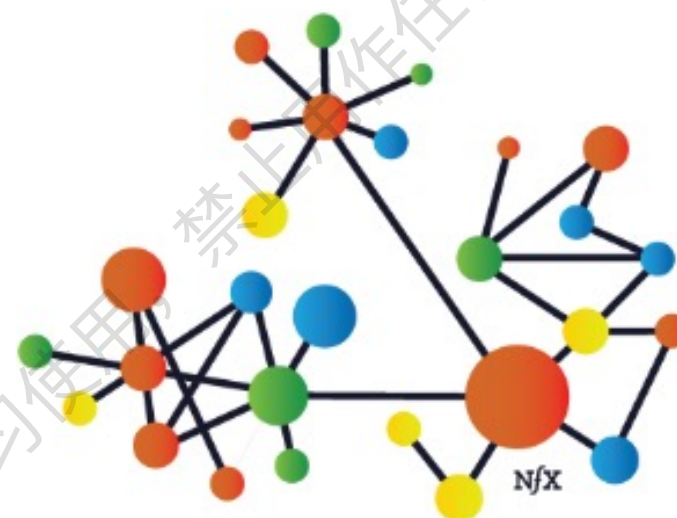
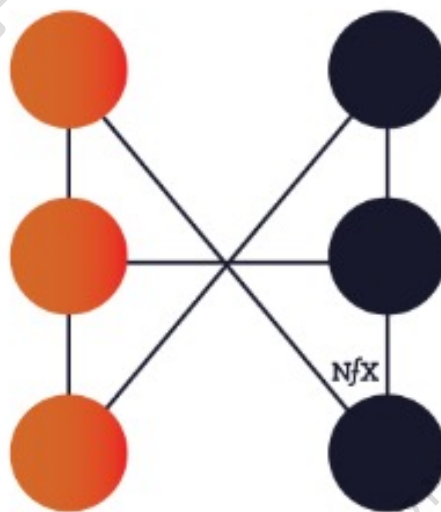
- 西方烹饪倾向于使用具有许多共同风味化合物的食材对
- 相比之下，东亚和南欧的菜系则避免使用含有相同风味成分的食谱



同质网络与异质网络 homogeneous & heterogeneous



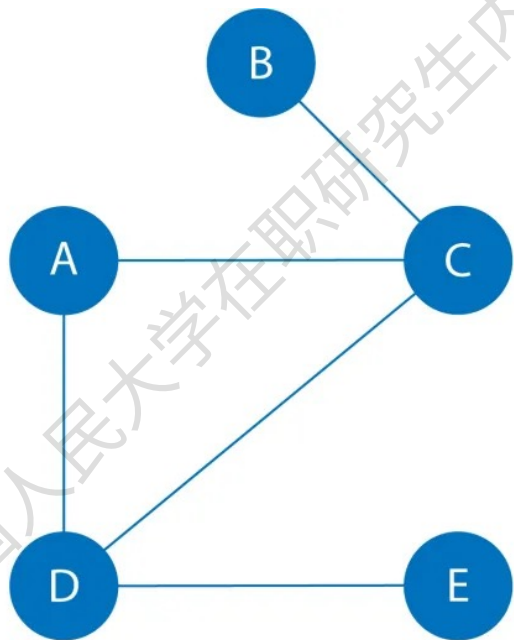
Homogeneous
Network



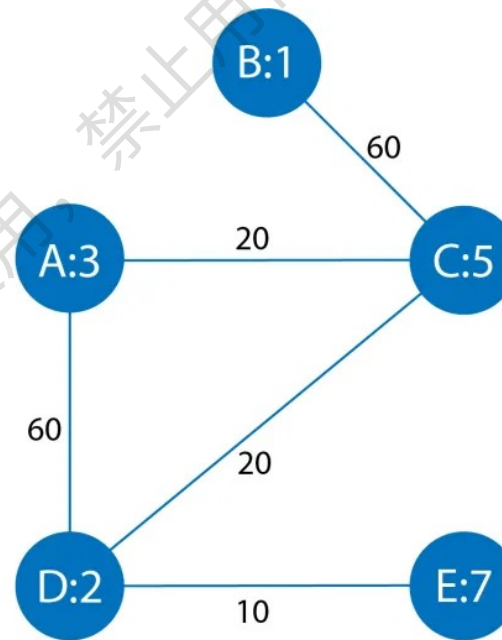
Heterogeneous
Networks

- 未加权图与加权图

- 未加权图（Unweighted Graphs）的节点和边上均没有权重。
- 加权图（Weighted Graphs），所加权重可以代表：成本、时间、距离、容量等，权重代表关系的强度。



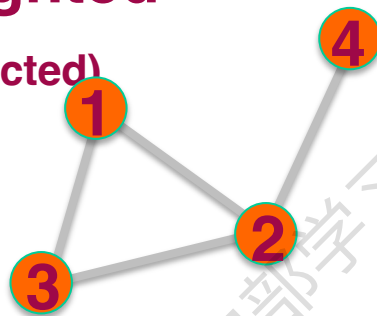
Unweighted



Weighted

Unweighted

(undirected)



$$A_{ij} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$A_{ii} = 0$$

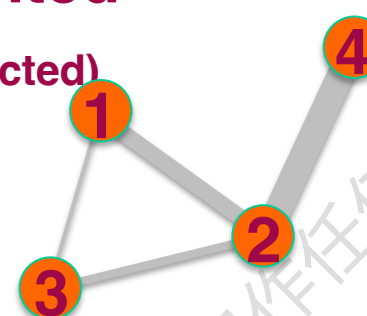
$$A_{ij} = A_{ji}$$

$$L = \frac{1}{2} \sum_{i,j=1}^N A_{ij} \quad \langle k \rangle = \frac{2L}{N}$$

protein-protein interactions, www

Weighted

(undirected)



$$A_{ij} = \begin{pmatrix} 0 & 2 & 0.5 & 0 \\ 2 & 0 & 1 & 4 \\ 0.5 & 1 & 0 & 0 \\ 0 & 4 & 0 & 0 \end{pmatrix}$$

$$A_{ii} = 0$$

$$A_{ij} = A_{ji}$$

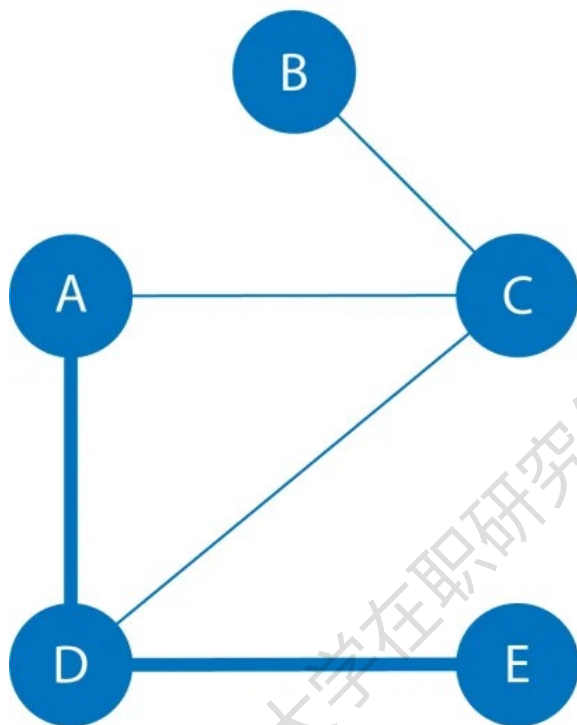
$$L = \frac{1}{2} \sum_{i,j=1}^N \text{nonzero}(A_{ij}) \quad \langle k \rangle = \frac{2L}{N}$$

Call Graph, metabolic networks

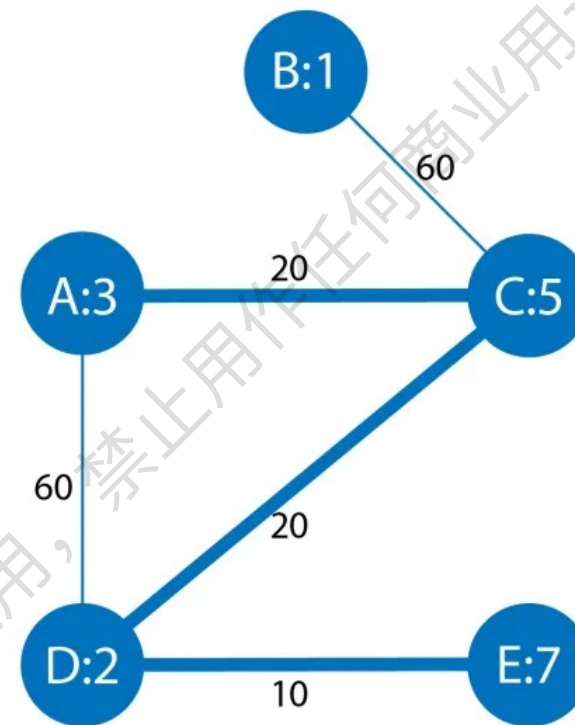
加权图应用



What's the Shortest Path
Between A and E?



Unweighted
A, D, E = 2 Hops



Weighted
A, C, D, E = 50

- 社会网络常见分类

- ✓ 复合型网络

from	to	好友关系	聊天关系	
A	B	1	1	
B	C	1	0	
C	A	0	1	
.....				

好友关系矩阵

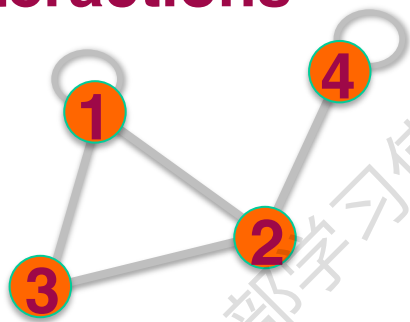
	A	B	C
A	-	1	0
B	1	-	1
C	0		-

聊天关系矩阵

	A	B	C
A	-	1	1
B	1	-	0
C	1		-

4 rows, 4 columns, 2 level

Self-interactions



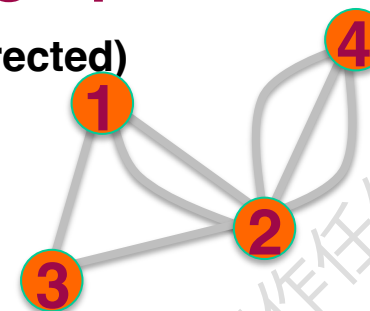
$$A_{ij} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

$$L = \frac{1}{2} \sum_{i,j=1, i \neq j}^N A_{ij} + \sum_{i=1}^N A_{ii} \quad ? \quad A_{ij} = A_{ji} \quad A_{ii} \neq 0$$

Protein interaction network, www

Multigraph

(undirected)



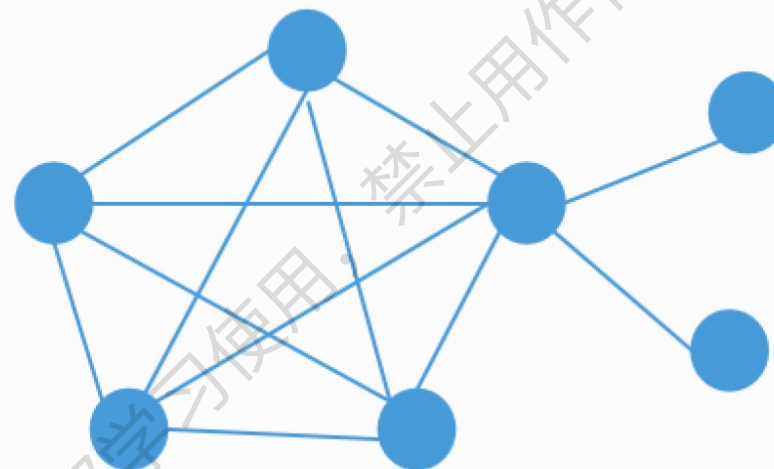
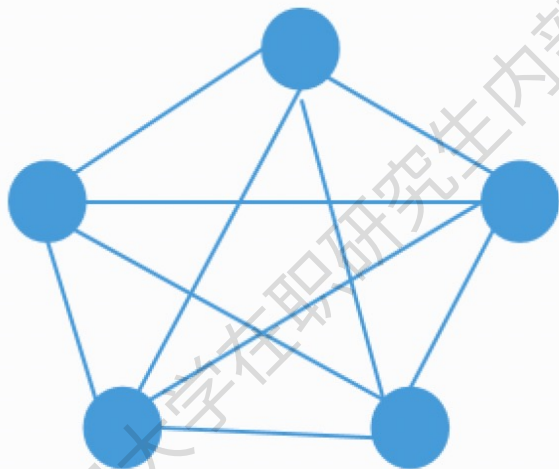
$$A_{ij} = \begin{pmatrix} 0 & 2 & 1 & 0 \\ 2 & 0 & 1 & 3 \\ 1 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \end{pmatrix}$$

$$L = \frac{1}{2} \sum_{i,j=1}^N \text{nonzero}(A_{ij}) \quad < k > = \frac{2L}{N} \quad A_{ii} = 0 \quad A_{ij} = A_{ji}$$

Social networks, collaboration networks

✓ 依据**成员紧密程度**分类：完备图 VS 非完备图

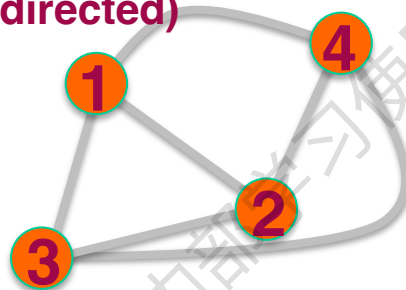
如果图中任意两个节点之间都有连线，则该图为完备图，否则为非完备图。



完备图的中心势为0（每个点都有相互联系，无所谓中心不中心），星型或辐射型的网络的中心势接近1

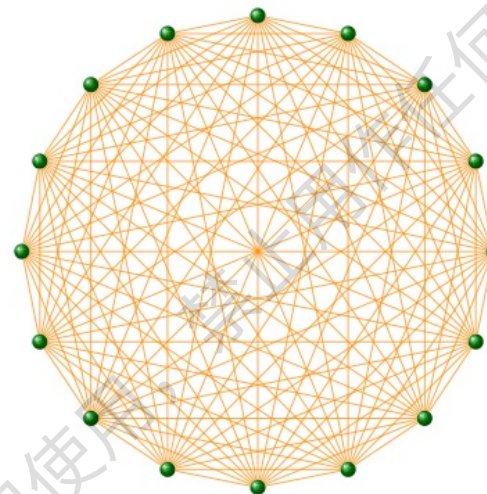
Complete Graph

(undirected)



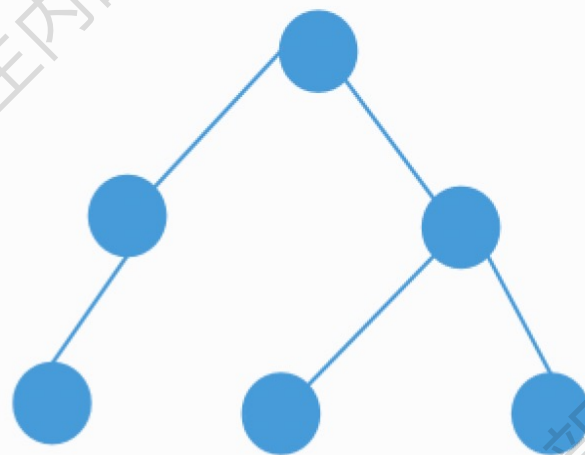
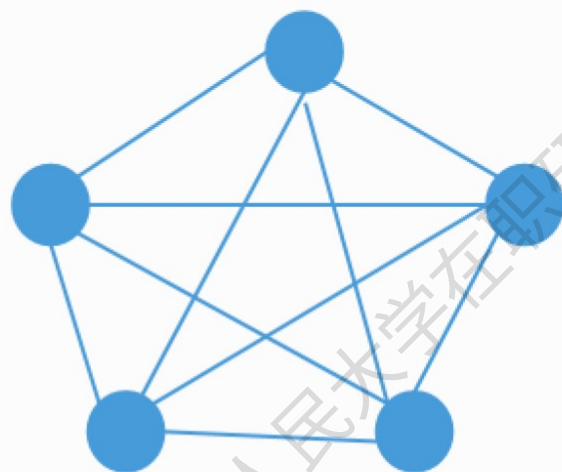
$$A_{ij} = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

$$A_{ii} = 0 \quad A_{i \neq j} = 1$$
$$L = L_{\max} = \frac{N(N-1)}{2} \quad \langle k \rangle = N-1$$



- “连通图” (connected graph) 与 “成分” (components)

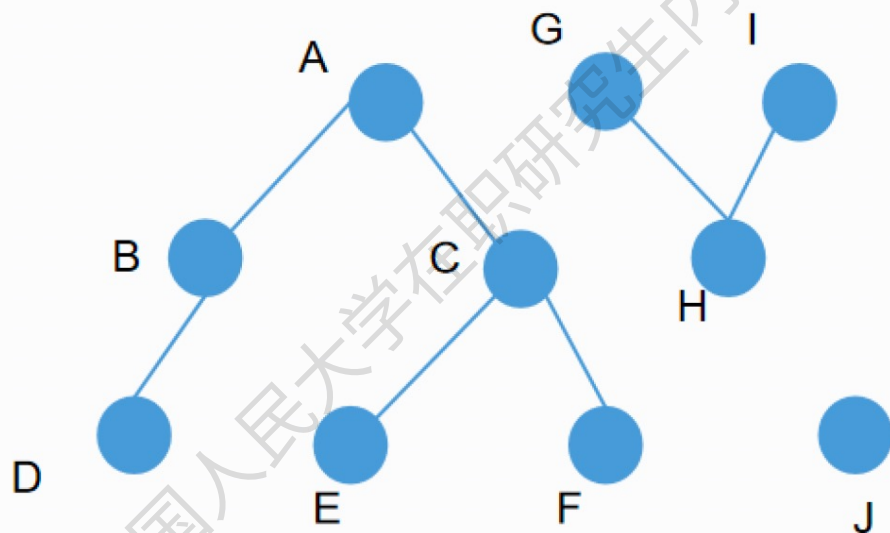
✓ 如果网络图中的任何两点之间都存在一个途径 (Path)，则称这两点是相互可达的，称该图时连通图 (connected graph)。也就是说，关联图中的任何两点之间都是可达的。



- 在有向图G中，如果两个顶点 v_i, v_j 间 ($v_i > v_j$) 有一条从 v_i 到 v_j 的有向路径，同时还有一条从 v_j 到 v_i 的有向路径，则称两个顶点**强连通(strongly connected)**。如果有向图G的每两个顶点都强连通，称G是一个**强连通图**。

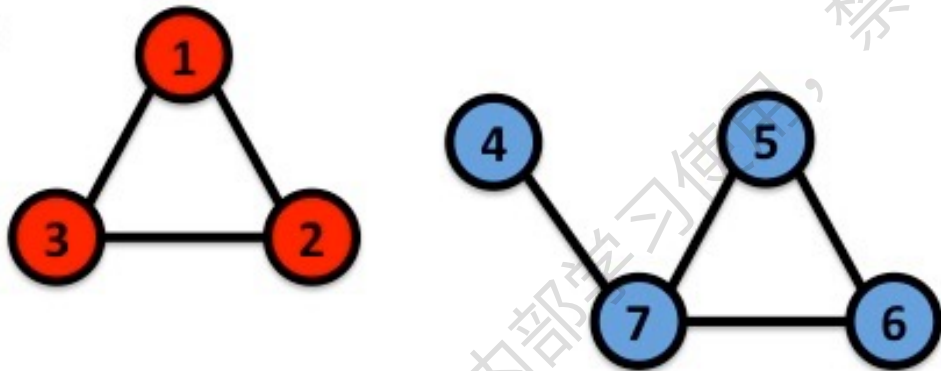
- “连通图” (connected graph) 与 “成分” (components)

- ✓ 如果一个图不是连通的，就称之为“不连通图”，其可以分为两个或者多个连通的子图，我们称之为连通子图。
- ✓ 一个图中的各个连通子图都叫做“成分” (components)。“成分”内部的任何点之间都存在途径。但是，成分内部的一点与任何外在于该成分的点之间都不存在任何途径。



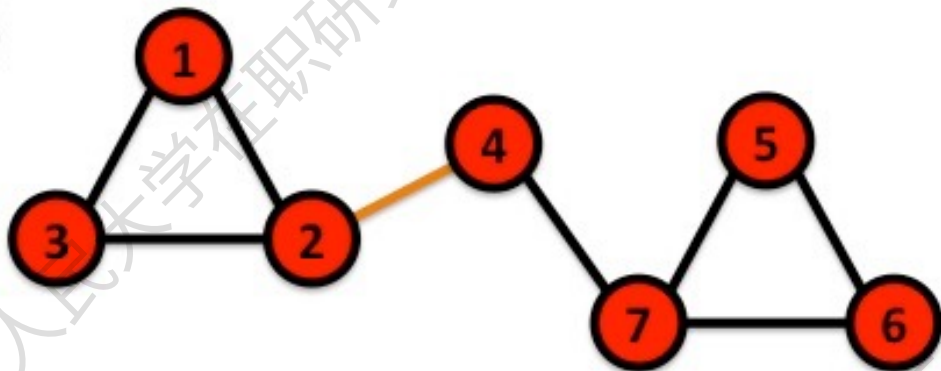
- $C1 = \{A, B, C, D, E, F\}$
- $C2 = \{G, H, I\}$
- $C3 = \{J\}$

(a)



$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}$$

(b)

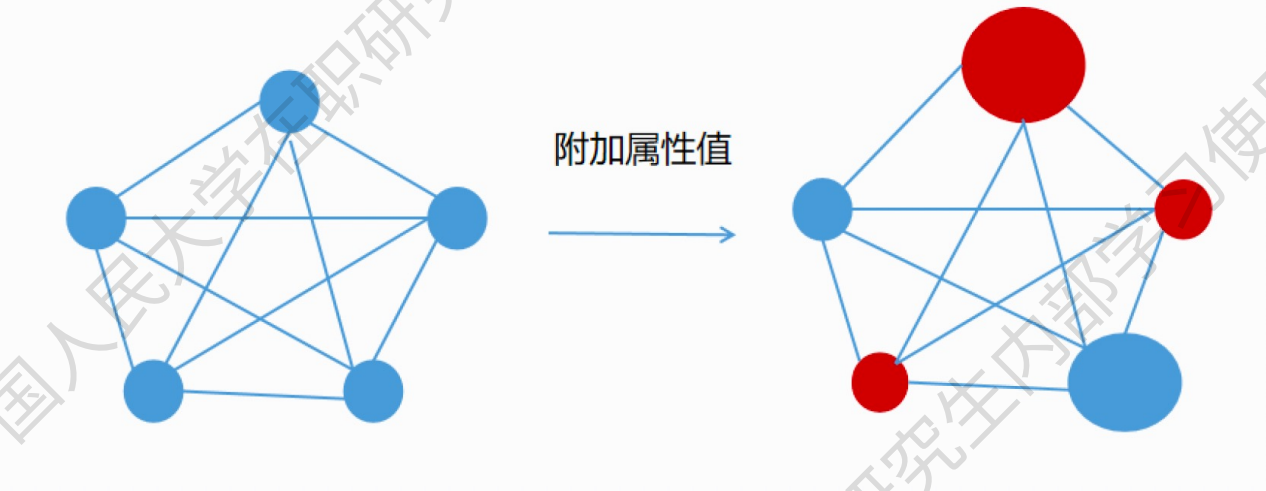


$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}$$

• 网络变量 + 属性变量

from	to	好友	聊天	
A	B	1	1	
B	C	1	0	
C	A	0	1	
.....				

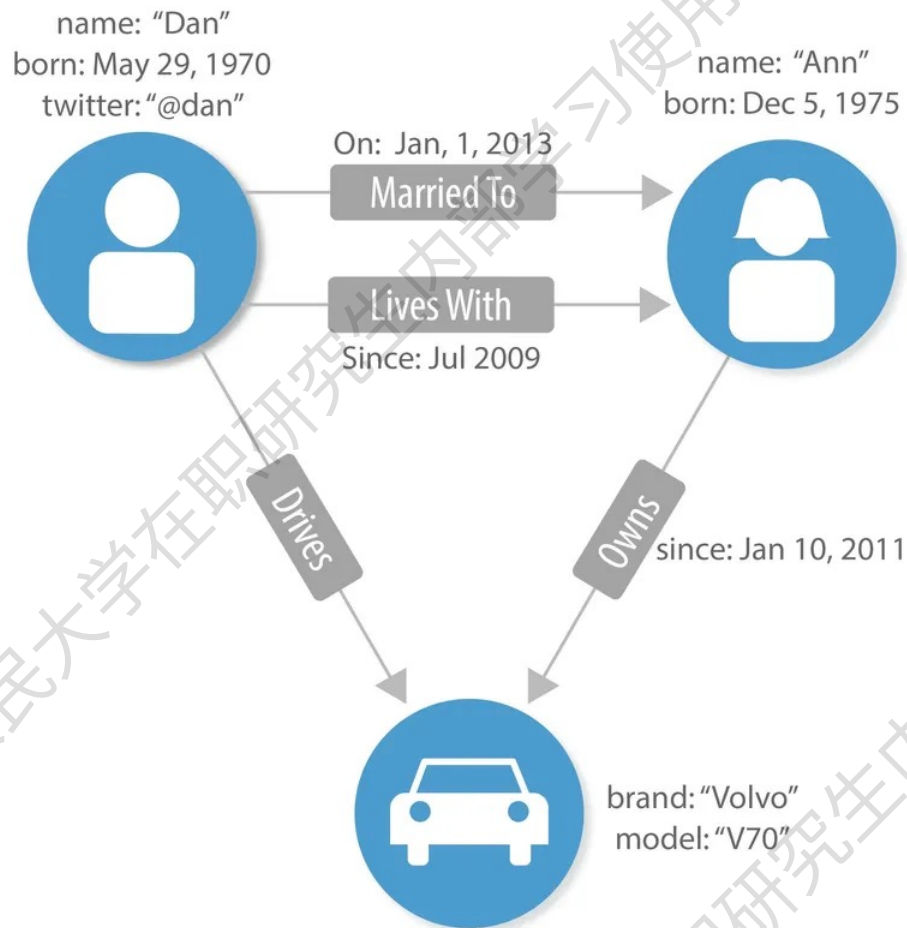
节点	年龄	性别	学历	
A	29	女	硕士	
B	30	男	本科	
C	31	女	博士	
.....				



一个社会网络例子



- 节点、关系、属性（Property）、标签等



Nodes

- Can have *labels* for classification
- Labels have native indexes

Relationships

- Relate nodes by *type* and direction

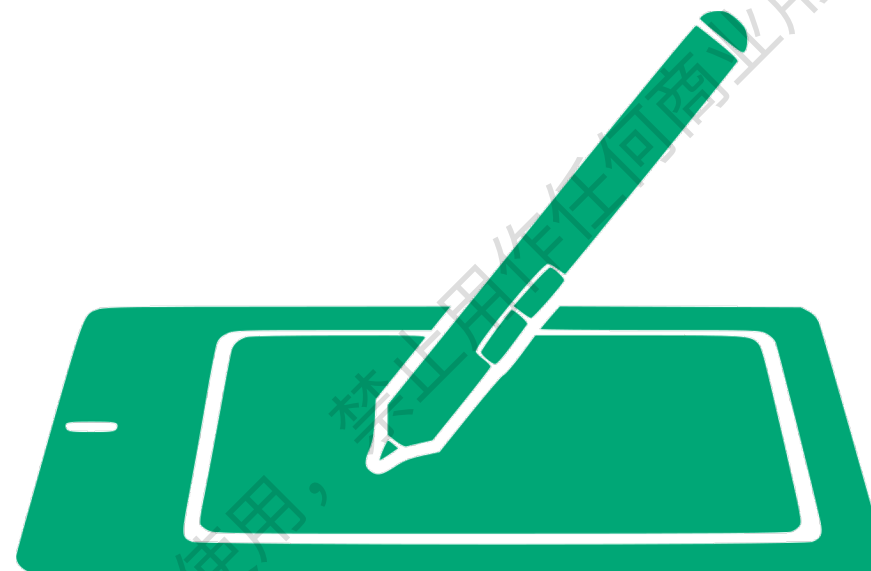
Properties

- *Attributes* of nodes & relationships
- Stored as name-value pairs
- Can have indexes and composite indexes

Draw your network



- 有向；无向
- 未加权；加权
- 同质；异质
- 复合型？
- 节点属性？
- 关系属性？



基础概念

社会网络类型

节点测度指标

路径搜索

社群发现

Gephi 实操

- 度 (degree)
- 中心性 (centrality)
 - 用于识别图中特定节点的角色及其对网络的影响。
 - 中心性算法能够帮助我们识别最重要的节点，例如可信度、可访问性、事物传播的速度等。

- 度、入度、出度

1

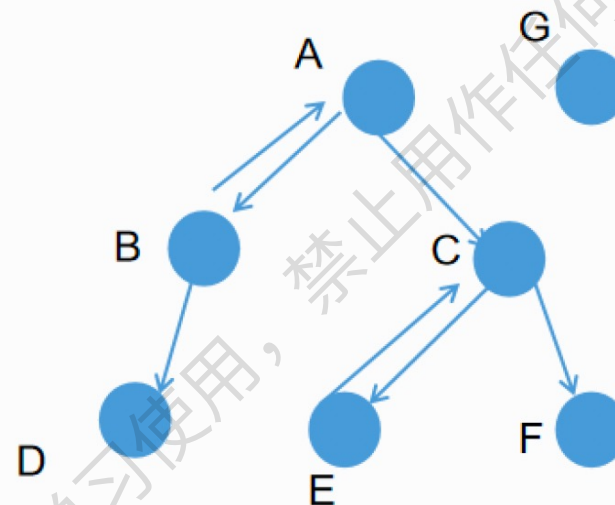
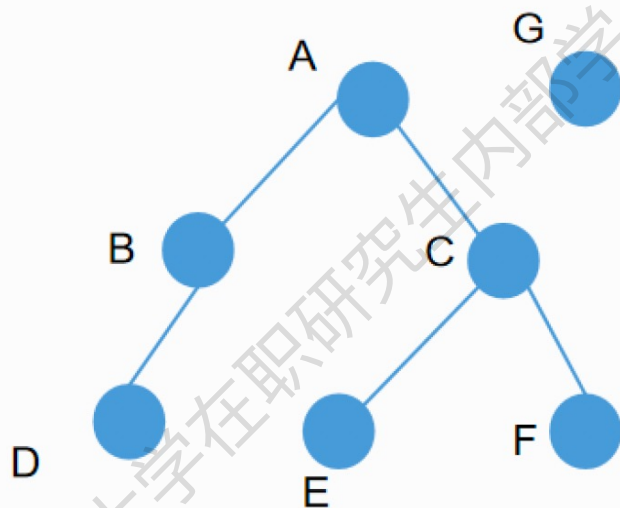
- “**度数**” (**degree**) : 与某节点直接相连的点称为该点的“邻点” (neighborhood) , 一个节点的邻点个数称为该点的“度数”也叫关联度 (degree of connection) 。 (度数为0称之为“孤立点” (isolate))

例如, 在一个社交网络中, 一个拥有更多 **degree** 的人 (节点) 更容易与人发生直接接触, 也更容易获得流感。

2

- “**点入度**” (**in-degree**) 和 “**点出度**” (**out-degree**) : 在有向网络中, 点入度指的是直接**指向该点**的点的总数; 点出度指的是**该点所直接**指向的点的总数。

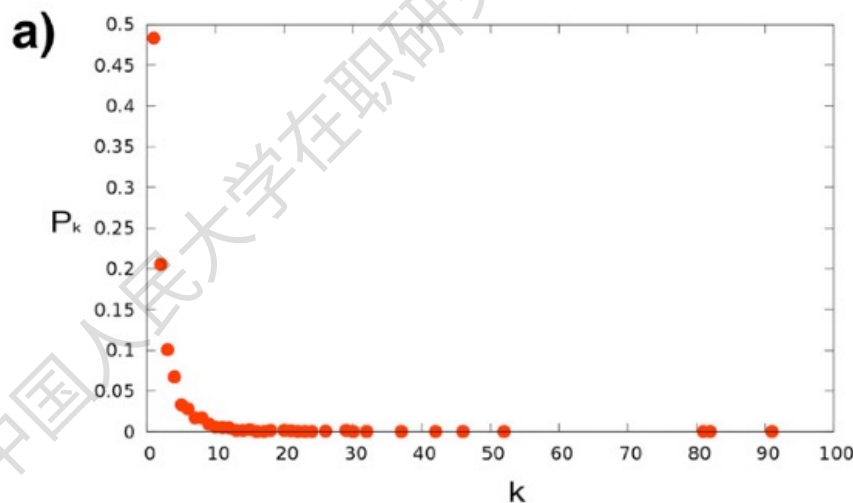
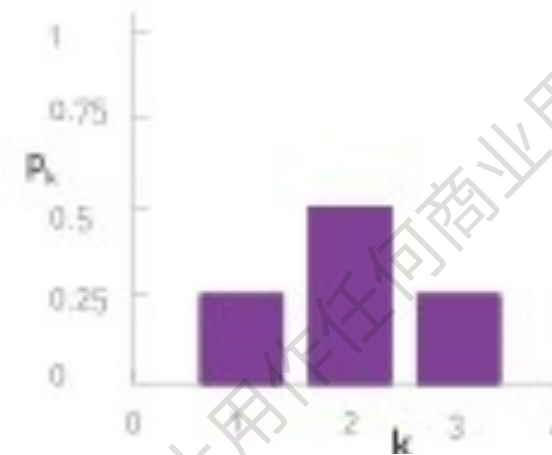
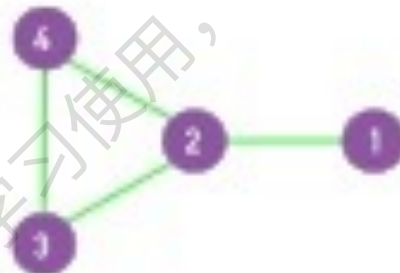
- 度、入度、出度



度分布 (Degree distribution)



$P(k)$: probability that a randomly chosen node has degree k



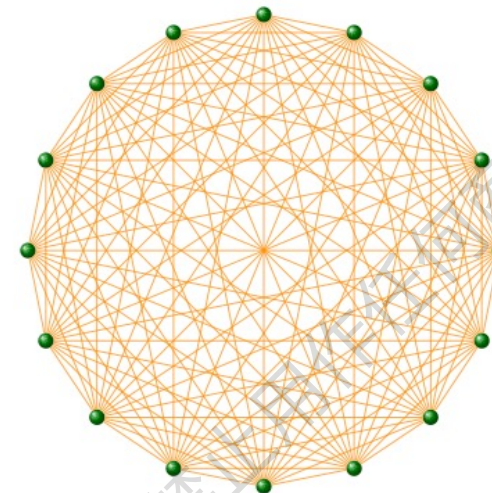
幂律分布

$$P(K=1) = 0.48$$

Real networks are sparse



The maximum number of links a network of N nodes can have is: $L_{\max} = \binom{N}{2} = \frac{N(N-1)}{2}$



A graph with degree $L=L_{\max}$ is called a **complete graph**, and its average degree is **$\langle k \rangle = N-1$**

Real networks are sparse



Most networks observed in real systems are sparse:

$$L \ll L_{\max}$$

or

$$\langle k \rangle \ll N-1.$$

WWW (ND Sample):	N=325,729;	$L=1.4 \cdot 10^6$	$L_{\max}=10^{12}$	$\langle k \rangle=4.51$
Protein (<i>S. Cerevisiae</i>):	N= 1,870;	$L=4,470$	$L_{\max}=10^7$	$\langle k \rangle=2.39$
Coauthorship (Math):	N= 70,975;	$L=2 \cdot 10^5$	$L_{\max}=3 \cdot 10^{10}$	$\langle k \rangle=3.9$
Movie Actors:	N=212,250;	$L=6 \cdot 10^6$	$L_{\max}=1.8 \cdot 10^{13}$	$\langle k \rangle=28.78$

(Source: Albert, Barabasi, RMP2002)

Real networks are sparse



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ADJACENCY MATRICES ARE SPARSE



酵母蛋白-蛋白相互作用网络的邻接矩阵，由2018个节点组成，每个节点代表一个酵母蛋白

- 中心性centrality

- ✓ 点度中心性 (degree centrality)：一个节点的度数。
- ✓ 中介中心性 (betweenness centrality)：如果一个节点位于其他节点的多条**最短路径**上，那么该节点具有较大的中介中心性。
- ✓ 接近中心性 (closeness centrality)：如果节点到图中其他节点的**最短距离都很小**，那么它的接近中心性就很高。相比中介中心性，接近中心性更接近几何上的中心位置。（平均值）
- ✓ 特征向量中心性 (eigenvector centrality)：一个节点的中心性是**相邻节点中心性**的函数。节点所连接的节点越重要，其自身中心性也就越高。

.....

- Degree Centrality（度中心性）——最简单的社会网络分析“算法”
- Degree 可以简单理解为一个节点的访问机会的大小。

例如，在一个社交网络中，一个拥有更多 **degree** 的人（节点）更容易与人发生直接接触，也更容易获得流感。

绝对中心度

无向图中，点的绝对中心度即为该点的度数。

有向图中

内中心度

点入度

外中心度

点出度

相对点度中心度

无向图： $C'_{RD}(x) = (x \text{ 的度数}) / (n-1)$

有向图： $C'_{RD}(x) = (x \text{ 的点入度数} + x \text{ 的点出度}) / (2n-2)$

- 中介中心性（Betweenness Centrality）是一种检测节点对图中信息或资源流的影响程度的方法。
- 它通常用于寻找连接图的两个部分的桥梁节点，系统重要“媒介”，它们掌握着资源或者信息的流动性。

- 中心性测量：中介中心度

绝对中介/中间中心度：

$$C_{ABi} = \sum_j^n \sum_k^n b_{jk}(i) \quad , \quad j \neq k \neq i, \text{ 且 } j < k$$

相对中介/中间中心度：

$$C_{RBi} = \frac{2C_{ABi}}{n^2 - 3n + 2} \quad , 0 \leq C_{RBi} \leq 1$$

- 假设点j和k之间存在 g_{jk} 条路径，
- 点j和k之间存在的经过点i的路径数用 $g_{jk}(i)$ 来表示，
- 点i能够控制此两点的交往的能力用 $b_{jk}(i)$ 来表示，
- 那么， $b_{jk}(i) = g_{jk}(i) / g_{jk}$ 。

$$C_n^m = \frac{A_n^m}{m!} = \frac{n!}{m!(n-m)!} = C_n^{n-m}$$

! - Factorial阶乘

- 中介中心性（Betweenness Centrality）计算

绝对中介/中间中心度:

$$C_{ABi} = \sum_j^n \sum_k^n b_{jk}(i), \quad j \neq k \neq i, \quad \text{且 } j < k$$

相对中介/中间中心度:

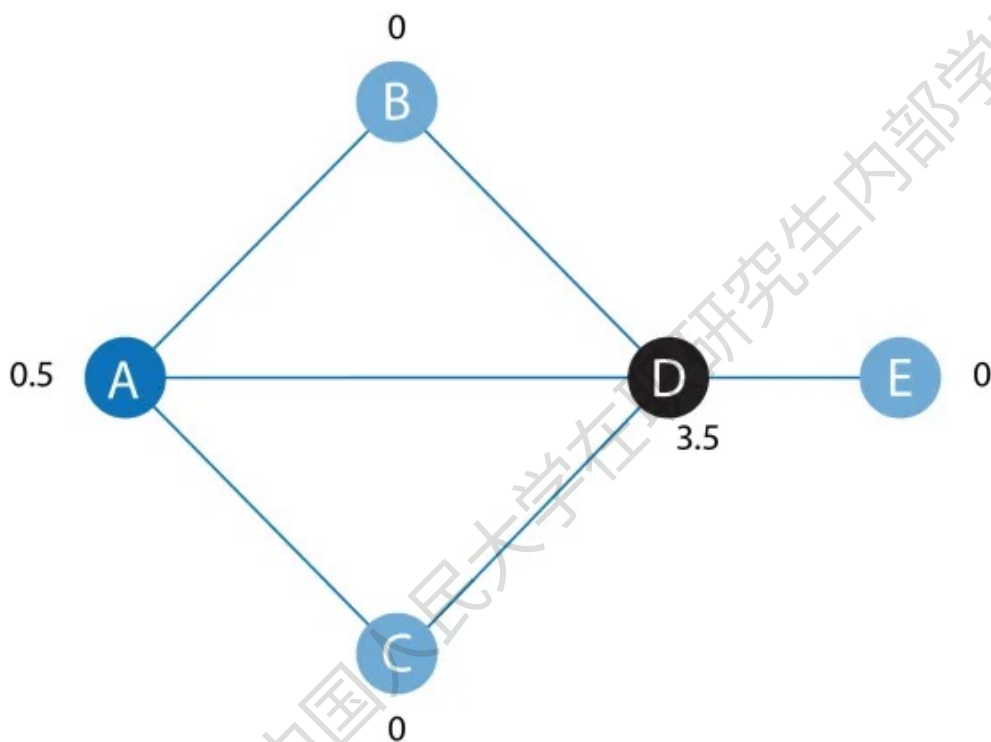
$$C_{RBi} = \frac{2C_{ABi}}{n^2 - 3n + 2}, \quad 0 \leq C_{RBi} \leq 1$$

- 假设点j和k之间存在 g_{jk} 条路径,
- 点j和k之间存在的经过点i的路径数用 $g_{jk}(i)$ 来表示,
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- 那么, $b_{jk}(i) = g_{jk}(i) / g_{jk}$ 。

$$C_n^m = \frac{A_n^m}{m!} = \frac{n!}{m!(n-m)!} = C_n^{n-m}$$

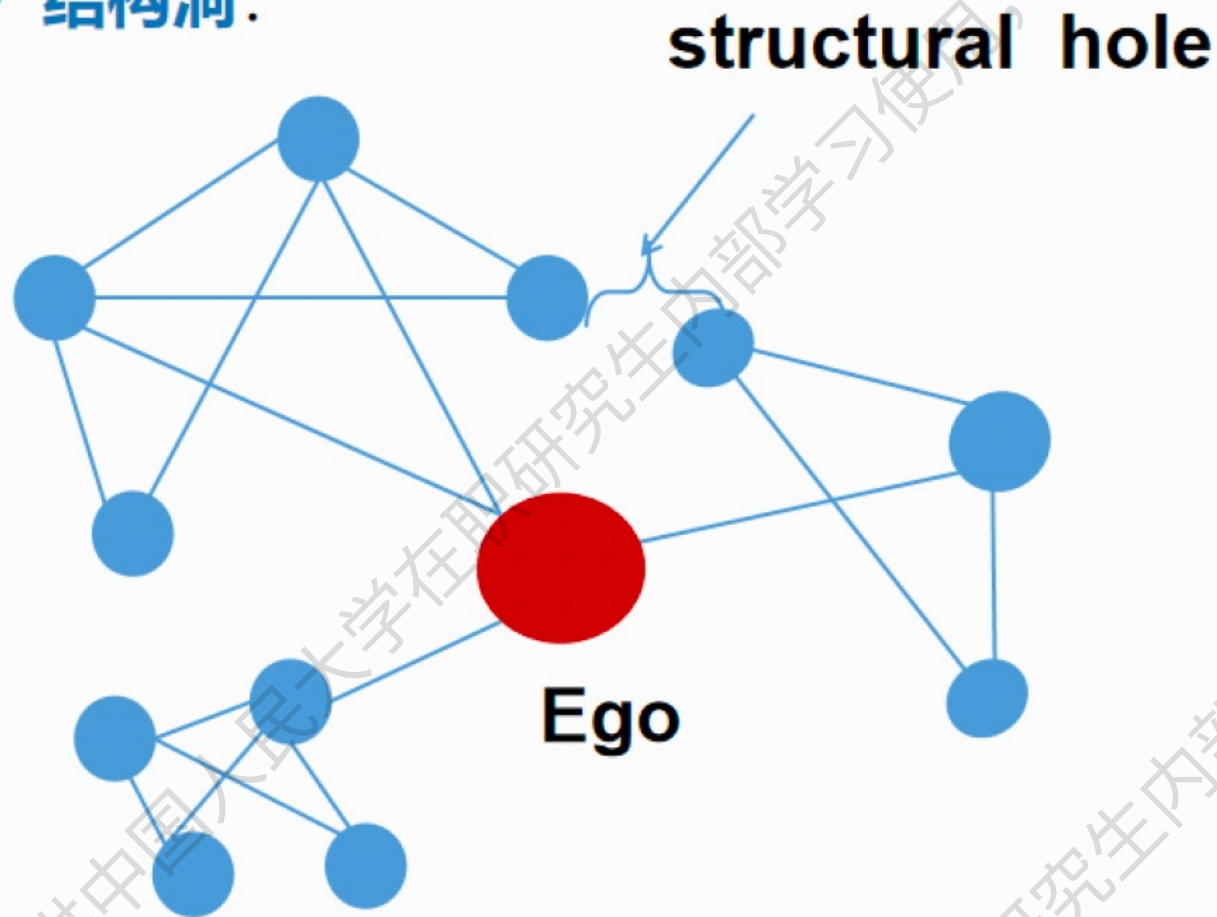
! - Factorial阶乘

- 中介中心性（Betweenness Centrality）计算



计算D节点的中介中心性

- 结构洞:



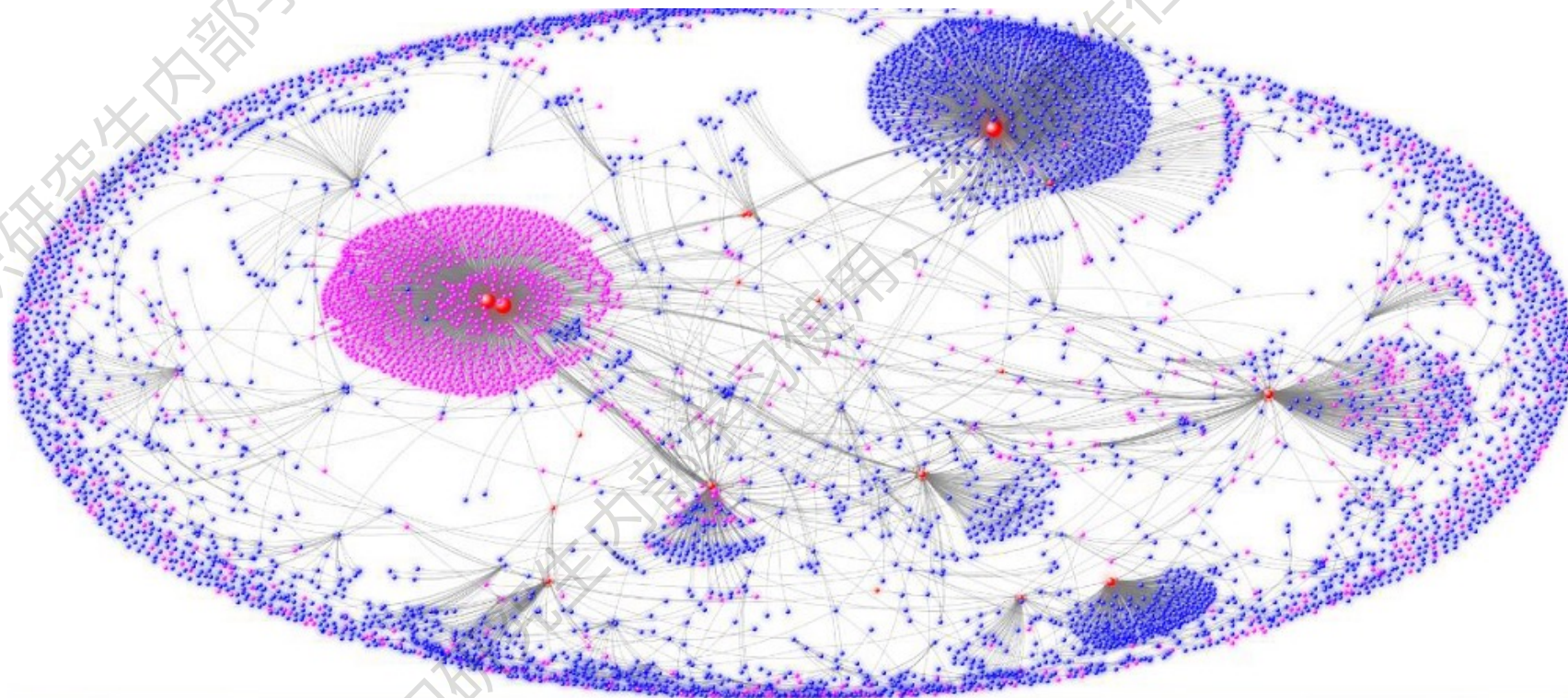
结构洞的存在使得Ego 处于**中间人**地位, 因而可以控制资源的传递等。

对于一个企业来说, 占据结构洞位置非常有利于非冗余、多元化信息的流动以及对信息流的控制, 从而也可能促进企业进行创新, 开发新产品。

节点的控制力

- 传播瓶颈（信息、疾病等）
- 一个在计算机科学学术世界和音乐世界的跨界者

基于社交网络的信息扩散分析



Closeness Centrality（紧密性中心性）计量一个节点到所有其他节点的紧密性（距离的倒数），一个拥有高紧密性中心性的节点拥有着到所有其他节点的距离最小值。

——适用于连通图

绝对接近中心度

$$C_{APi}^{-1} = \sum_{j=1}^n d_{ij}$$

(d_{ij} 为点*i*和*j*之间的捷径距离/测地线)

相对接近中心度

$$C_{RPi}^{-1} = \frac{C_{APi}^{-1}}{n - 1}$$

(n 为网络节点数)

网络中传播信息最快的节点

非连通图？

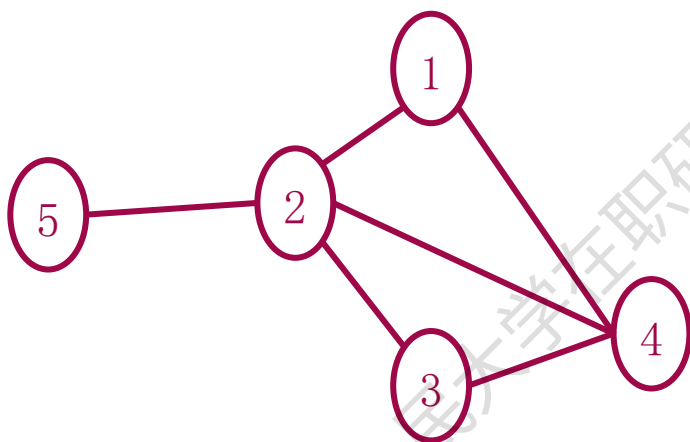
调和中心性 (Harmonic Centrality)

调和中心性提出的“平均最短距离”则是对这些最短距离的倒数求和，这使它可以处理非连通图中会出现的无限值。

- 特征向量中心性（eigenvector centrality）度量的是节点影响的传递。
- 来自高分值邻居节点对节点分值的贡献大于来自低分值邻居节点，节点有高分值意味着它连接到许多高分值节点。
- 特征向量中心性强调节点所处的周围环境，例如在疾病传播中，特征向量中心性较大的节点距离传染源更近的可能性更大，需要格外防范

- 特征向量中心性 (eigenvector centrality) 计算

对邻接矩阵求特征值，绝对值最大的特征值称为主特征值，主特征值对应的特征向量即为主特征向量，它包含矩阵最多的信息。



$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} \lambda_1 &= 2.686 \\ \lambda_2 &= -1.749 \\ \lambda_3 &= -1.271 \\ \lambda_4 &= 0.335 \\ \lambda_5 &= 0 \end{aligned}$$

Eigenvalues

$|\lambda_1|$ is the largest, so λ_1 is the dominate eigenvalue

$$x_1 = \begin{bmatrix} 1.899 \\ 2.686 \\ 1.899 \\ 2.414 \\ 1 \end{bmatrix}$$

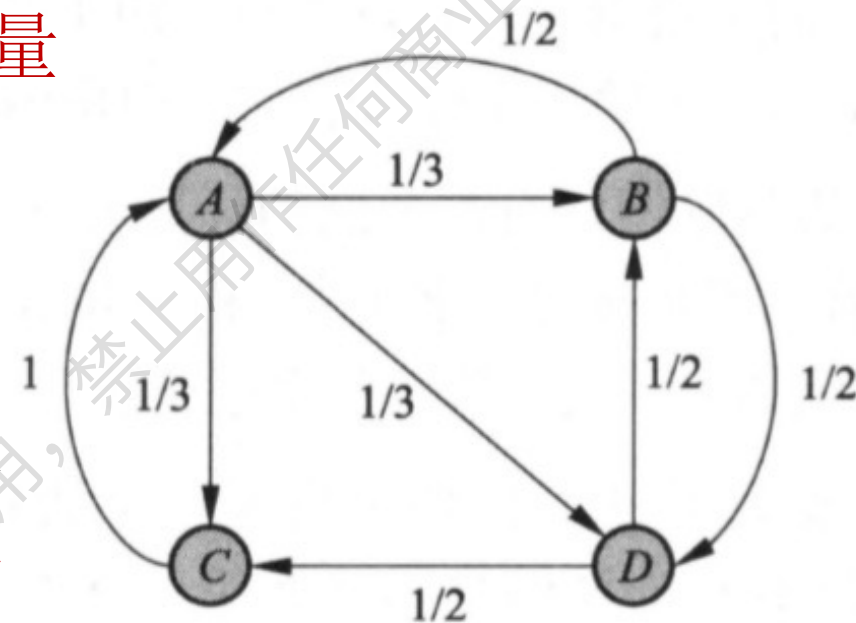
Dominate Eigenvector

2号节点特征向量中心性最大，5号节点最小

- PageRank

- Google 著名的 PageRank 算法是特征向量中心性算法的一个变种，用来进行网页排名。

- 假设互联网是一个有向图，在其基础上定义随机游走模型，即一阶马尔可夫链，表示网页浏览者在互联网上随机浏览网页的过程。



一个网页，如果指向该网页的超链接越多，如果指向该网页的PageRank值越高，随机跳转到该网页的概率也就越高，该网页的PageRank值就越高，这个网页也就越重要。

- PageRank 算法已经不仅限于网页排名。例如：

□寻找最重要的基因

我们要寻找的基因可能不是与生物功能联系最多的基因，而是与最重要功能有紧密联系的基因；

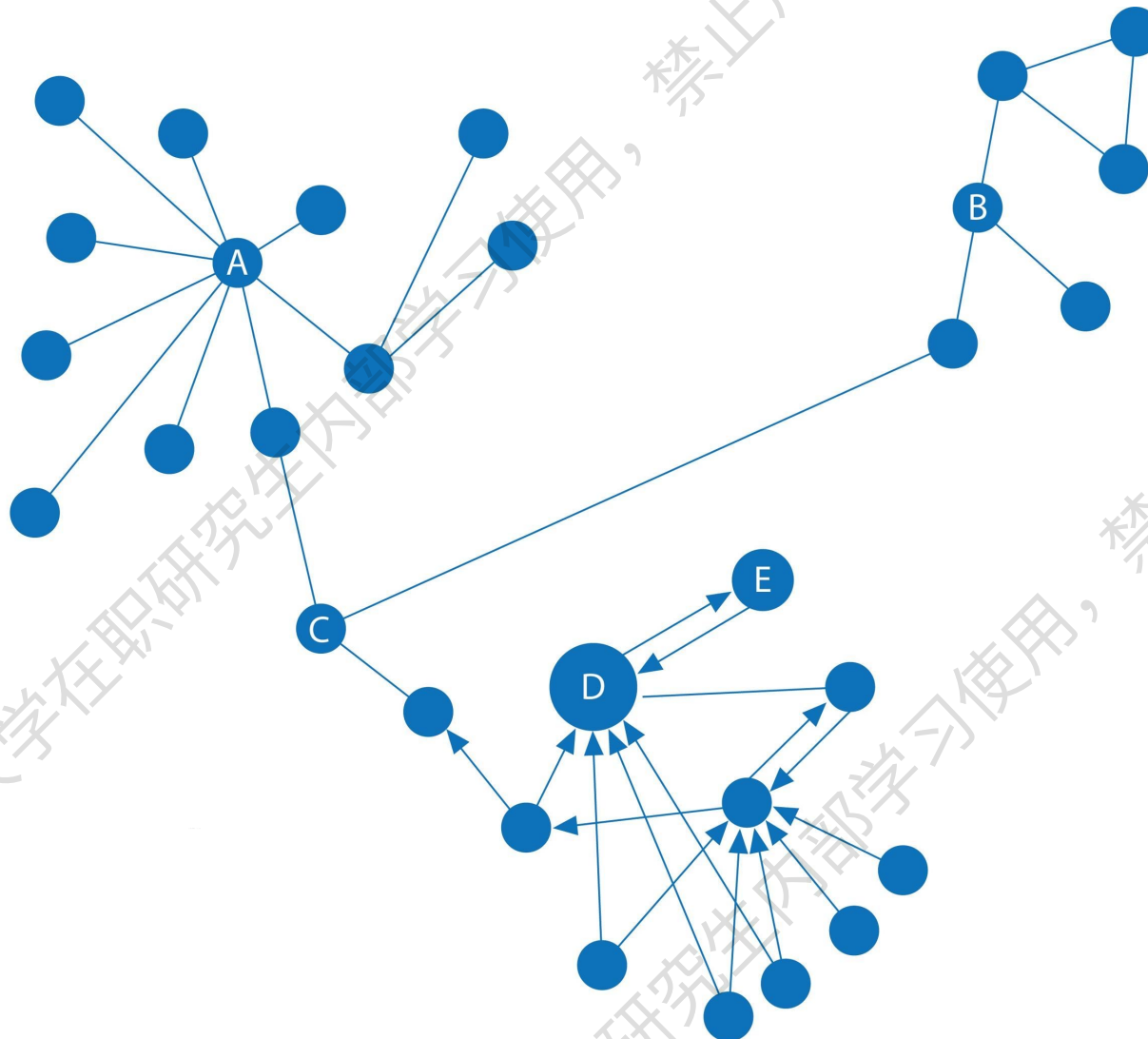
□who to follow service at twitter

Twitter使用个性化的 PageRank 算法向用户推荐他们可能希望关注的其他帐户。该算法通过兴趣和其他的关系连接，为用户展示感兴趣的其他用户；

□交通流量预测

使用 PageRank 算法计算人们在每条街道上停车或结束行程的可能性

节点度量



- 度中心性
- 中介中心性
- 接近中心性
- PageRank

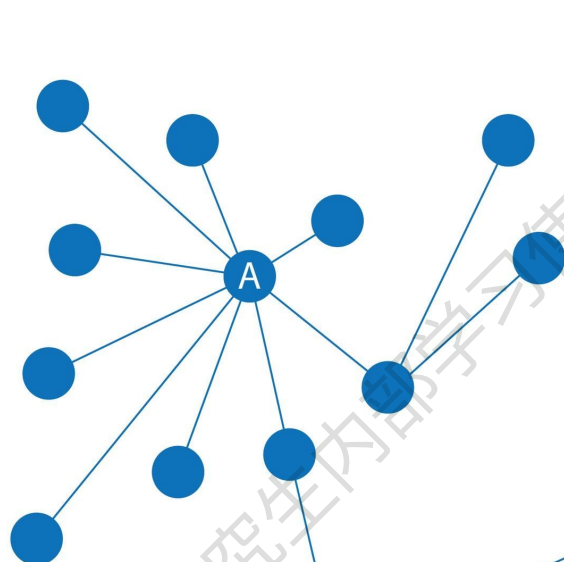
Source: <https://zhuanlan.zhihu.com/p/64984300>

节点度量

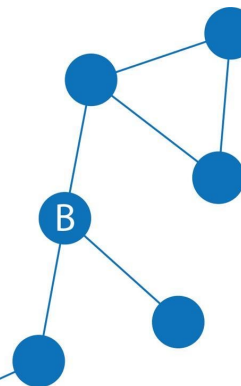


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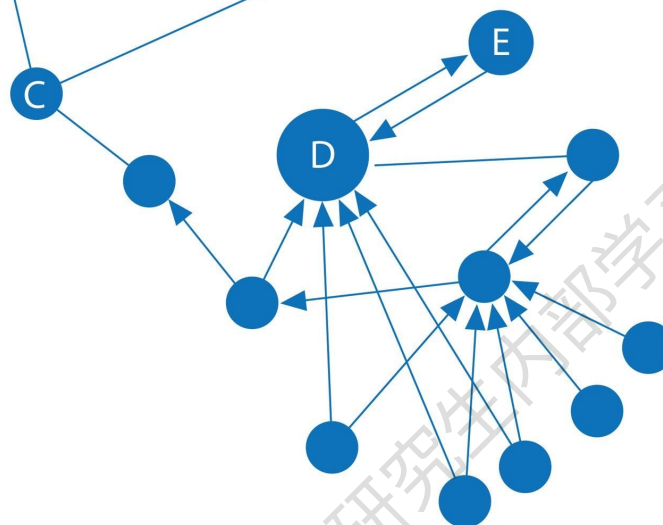
Degree
Number of connections?
"A" has a high degree



Closeness
Which node can most easily reach all other nodes in a graph or subgraph?
"B" is closest with the fewest hops in its subgraph



Betweenness
Which node has the most control over flow between nodes and groups?
"C" is a bridge



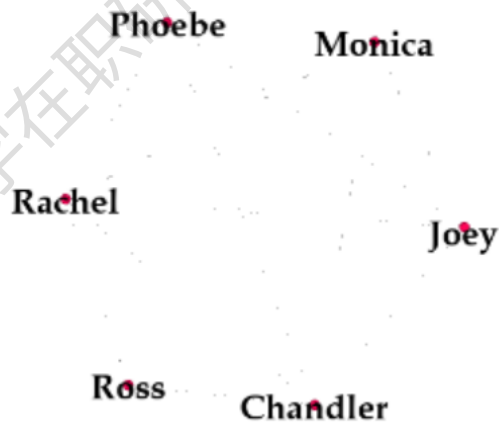
PageRank
Which node is the most important?
"D" is foremost based on number & weighting of in-links
"E" is next, due to the influence of D's link

Source: <https://zhuanlan.zhihu.com/p/64984300>

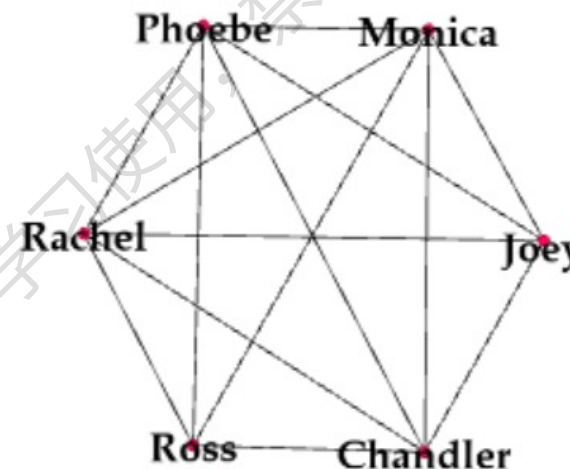
- 密度 (Density)
- 聚集系数 (Clustering coefficient)

- ✓ 衡量网络中节点间的边的密集程度
- ✓ 定义：网络中实际存在的边数与可容纳的边数上限的比值
- ✓ 举例：

空网络密度为0



全网络密度为1



聚集系数 (clustering coefficient)



✓ 衡量节点的好友之间有多大可能性是好友

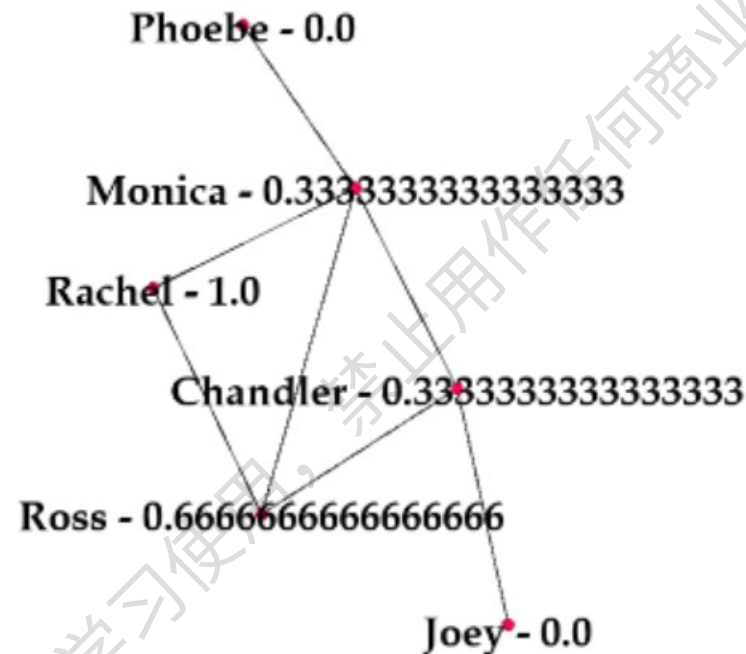
✓ 定义：节点的相邻节点间，边数

与所有潜在可能边数的比例

✓ 举例：Ross有三个好友，三个人的

潜在可能连接是3，实际连接是2，

Ross的聚集系数为 $2/3$ 。



网络聚集系数：节点聚集系数均值

基础概念

社会网络类型

节点测度指标

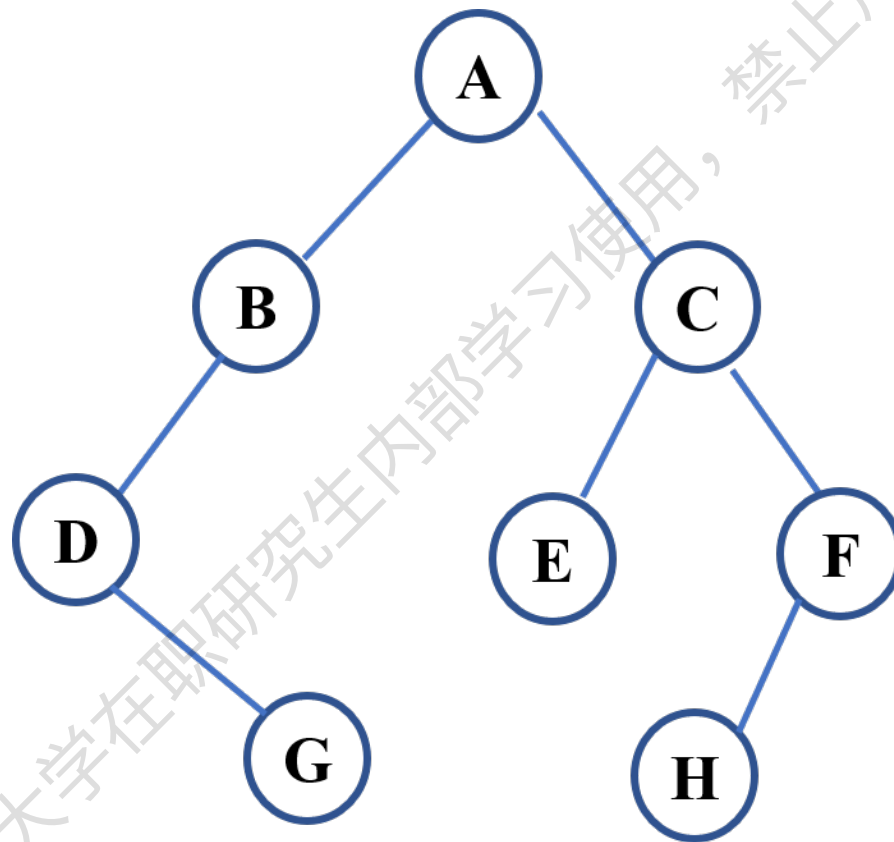
路径搜索

社群发现

Gephi 实操

- 图搜索（**Graph search**）
 - 探索一个图，用于一般发现或显式搜索，通常是许多其他类型分析的第一步。
- 路径搜索（**Pathfinding**）
 - 探索节点之间的路径，这些路径从一个节点开始，遍历关系，直到到达目的地。
 - 路径搜索算法识别最优路径，用于物流规划，最低成本呼叫或者IP路由问题，以及游戏模拟等。

- 图搜索算法包括**广度优先搜索**（**Breadth First Search**，简称 **BFS**）和**深度优先搜索**（**Depth First Search**，简称 **DFS**），它们是遍历图的基础
 - ▣ **BFS** 从选定的节点出发，优先访问所有一度关系的节点之后再继续访问二度关系节点，以此类推。
 - ▣ **DFS** 从选定的节点出发，选择任一邻居之后，尽可能的沿着边遍历下去，直到不能前进之后再回溯。



DFS: A-B-D-G-C-E-F-H

BFS: A-B-C-D-E-F-G-H

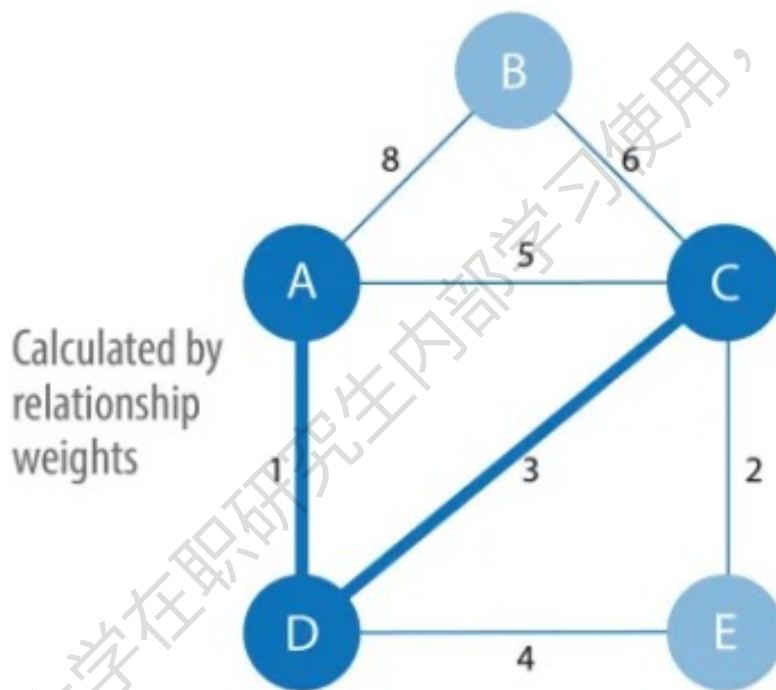
很少真正需要使用 BFS 和 DFS，这两个图搜索算法更多地作为底层算法支持其他图算法

- 最短路径（**Shortest Paths**）算法计算给定的两个节点之间最短（最小权重和）的路径，可以计算两点之间成本最低的路线。
 - 导航：谷歌、百度、高德地图均提供了导航功能，它们就使用了最短路径算法
 - 疾病传播：最短距离→有效距离。

最短路径shortest path



- 图



Shortest Path

Shortest path between 2 nodes (A to C shown)

(A, B) = 8

(A, C) = 4 via D

(A, D) = 1

(A, E) = 5 via D

(B, C) = 6

(B, D) = 9 via A or C

And so on...

All-Pairs Shortest Paths

Optimized calculations for shortest paths from all nodes to all other nodes

点之间最短的路线。

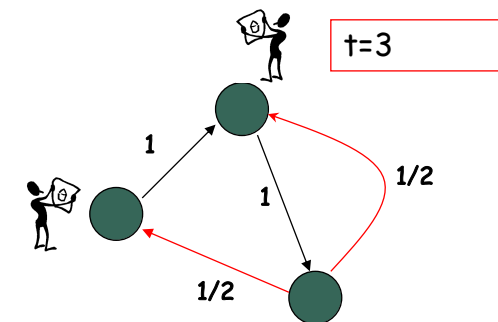
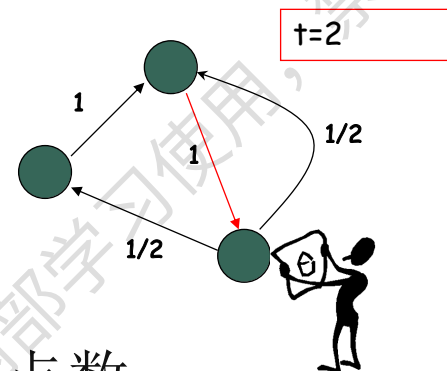
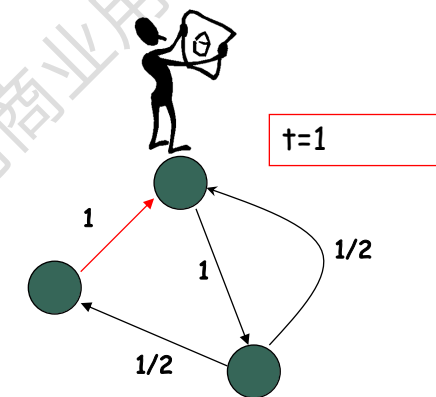
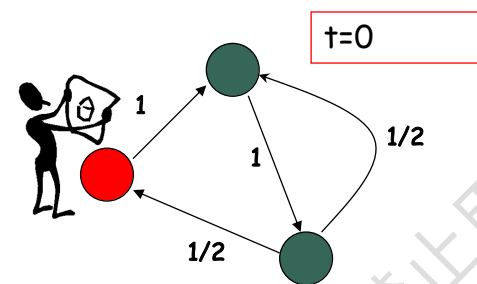
它们就使

好友，并列

随机游走Random walk

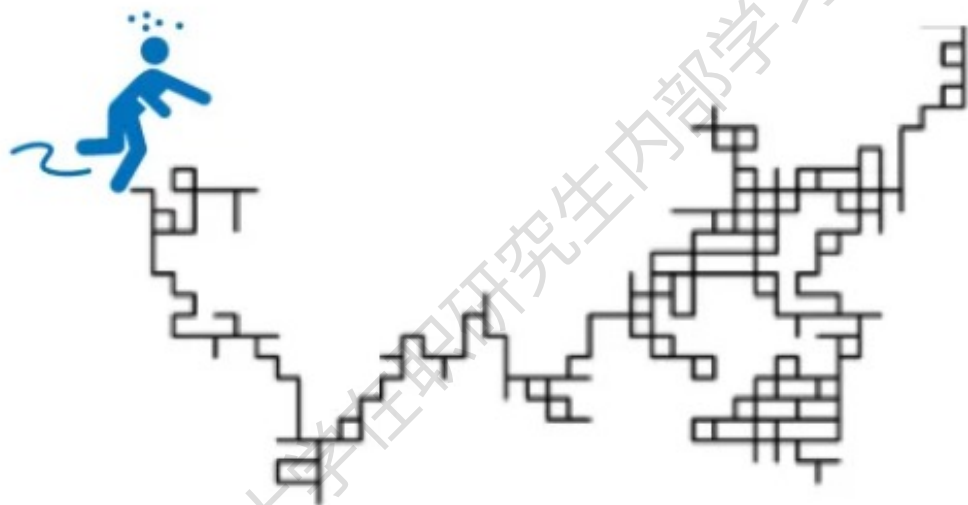


- 随机游走（Random Walk）算法从图上获得一条随机的路径
 - 从一个节点开始，随机沿着一条边正向或者反向寻找到它的邻居，以此类推，直到达到设置的路径长度



随机游走算法一般用于随机生成一组相关的节点数据，作为后续数据处理或者其他算法使用。

Random Pathfinding Algorithm



Random Walk

Provides a set of random, connected nodes by following any relationship, selected somewhat randomly.

Also called the drunkard's walk.

基础概念

社会网络类型

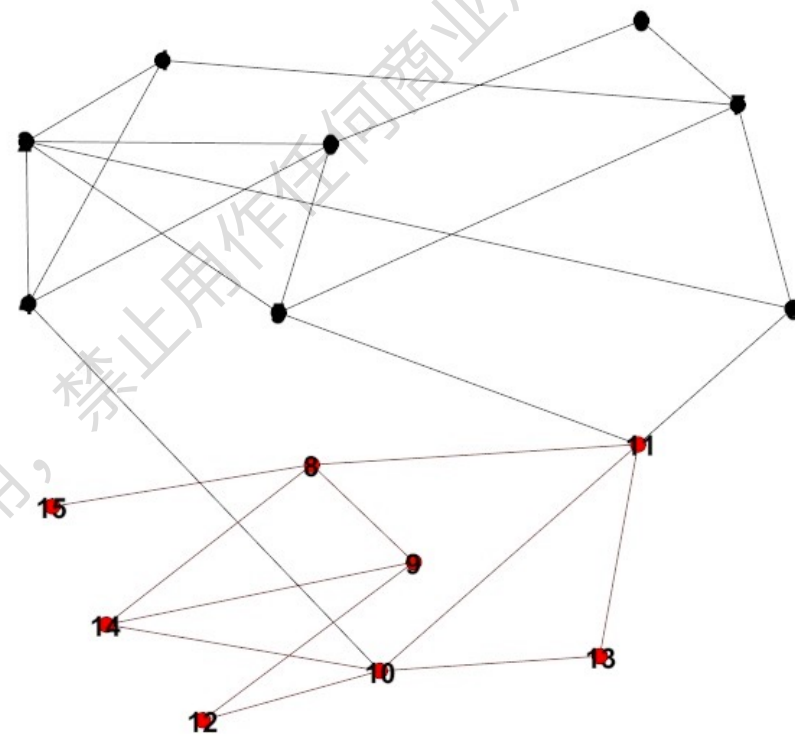
节点测度指标

路径搜索

社群发现

Gephi 实操

- 在社会网络中，有的用户之间的连接较为**紧密**，有的用户之间的连接关系较为**稀疏**。
- 其中连接较为紧密的部分可以被看成一个**社群/社区（community）**，其内部的节点之间有较为紧密的连接，而在两个社区间则相对连接较为稀疏。



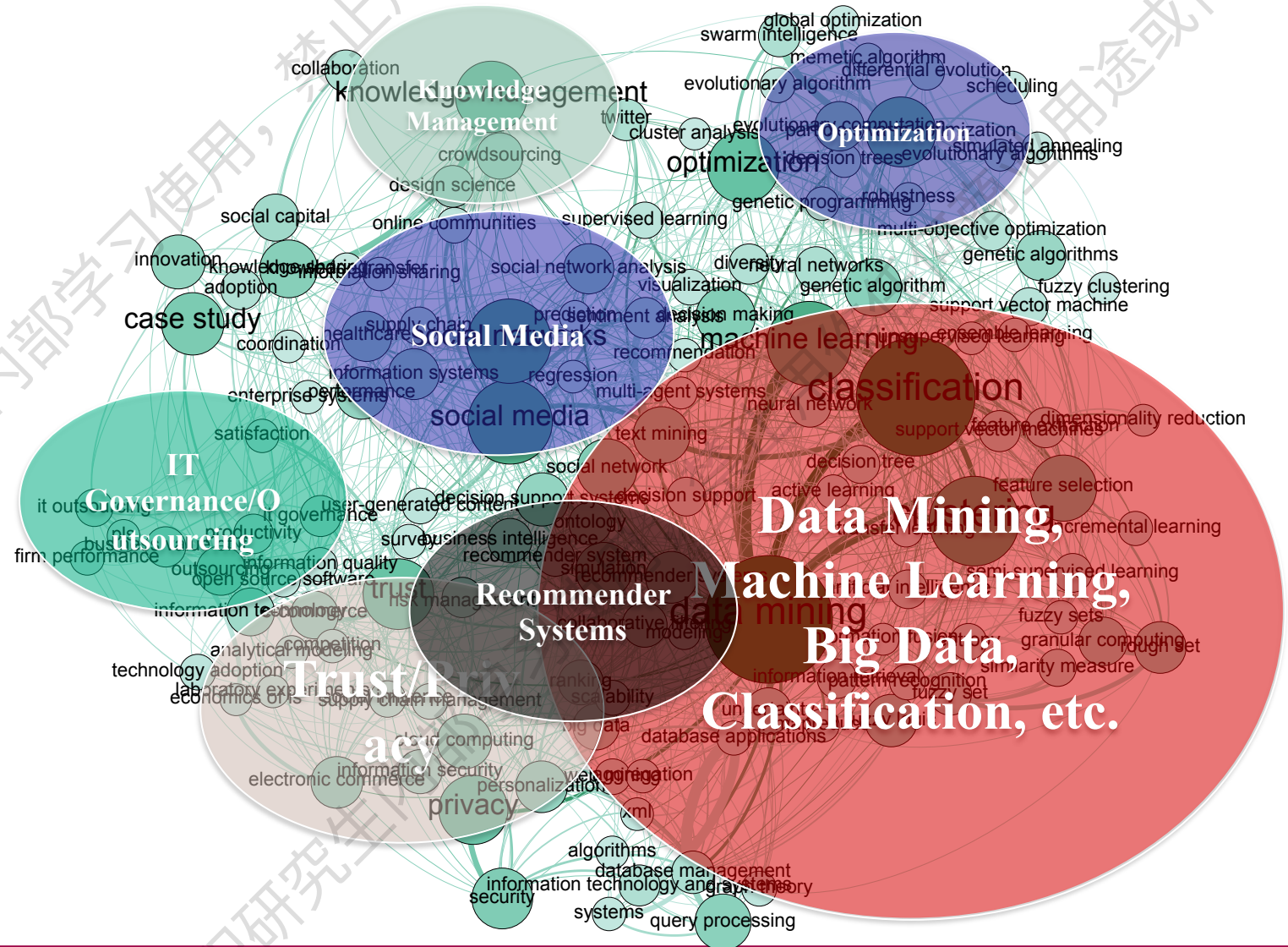
社群发现算法（Community Detection Algorithms）有助于发现社群中群体行为或者偏好，寻找嵌套关系，或者成为其他分析的前序步骤

社群发现-word association network

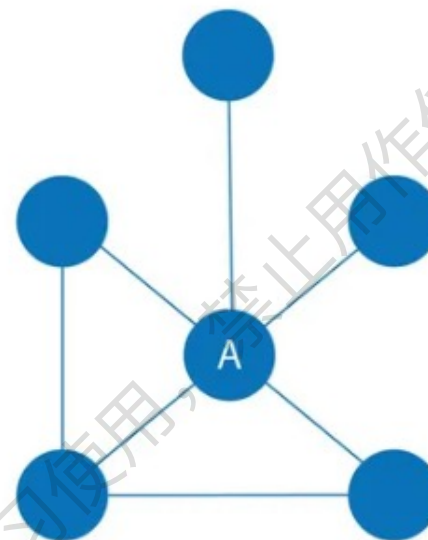


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- IS knowledge graph



- **三角计数 (Triangle Count)** 计算图中由节点组成的三角形的数量，要求任意两个节点间有边（关系）连接。
- **聚类系数 (Clustering Coefficient)** 算法的目标是测量一个组的聚类紧密程度。
 - 该算法计算网络中三角形的数量，与可能的关系的比率。聚类系数为 1 表示这个组内任意两个节点之间有边相连。



Triangle Count

The number of triangles that pass through a node. A has 2 triangles.

Clustering Coefficient

The probability that the neighbors of a node are connected to each other.

A has a 0.2 CC. Any 2 nodes connected to A have a 20% chance of being connected to each other.

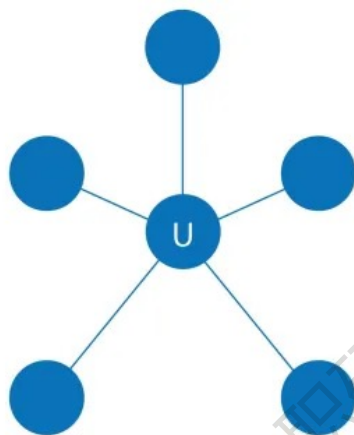
These measures can be counted/normalized globally

聚类系数 (Clustering Coefficient)



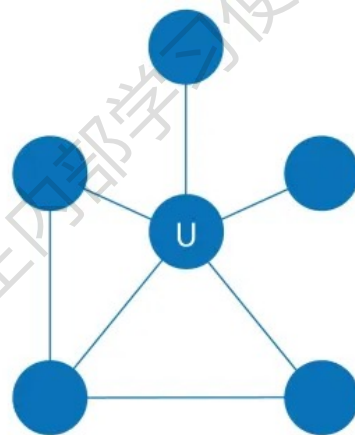
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- 局部聚类系数计算一个节点的邻居之间的紧密程度



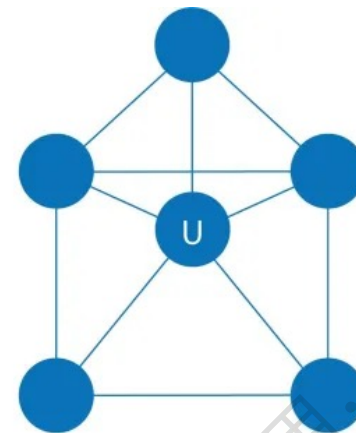
Triangles = 0
Clustering Coefficient = 0

$$CC(u) = \frac{0(2)}{5(5-1)}$$



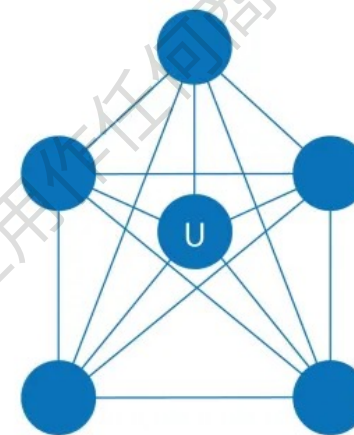
Triangles = 2
Clustering Coefficient = 0.2

$$CC(u) = \frac{2(2)}{5(5-1)}$$



Triangles = 6
Clustering Coefficient = 0.6

$$CC(u) = \frac{6(2)}{5(5-1)}$$



Triangles = 10
Clustering Coefficient = 1

$$CC(u) = \frac{10(2)}{5(5-1)}$$

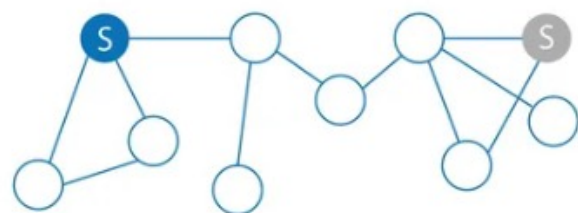
- 全局聚类系数是局部聚类系数的归一化求和。

- 强关联部件（**Strongly Connected Components**，简称 **SCC**）算法寻找有向图内的一组节点，每组节点可以通过关系互相访问。
 -
- 每组节点内部不需要直接相连，只要通过路径访问即可。
- 关联类算法
 - 标签传播算法（**Label Propagation Algorithm**，简称 **LPA**）
 - **Louvain Modularity Algorithm**

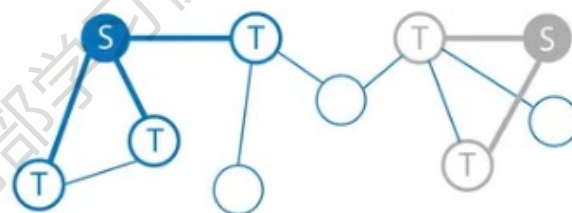
- 标签传播算法（Label Propagation Algorithm，简称 LPA），由 Raghavan 等人于 2007 年提出，是一个在图中快速发现社群的算法。
- 在 LPA 算法中，节点的标签完全由它的直接邻居决定。
- 形象地理解算法的传播过程：
 - ❑ 当标签在紧密联系的区域，传播非常快，但到了稀疏连接的区域，传播速度就会下降。
 - ❑ 当出现一个节点属于多个社群时，算法会使用该节点邻居的标签与权重，决定最终的标签。
 - ❑ 传播结束后，拥有同样标签的节点被视为在同一群组中。

Label Propagation - PUSH

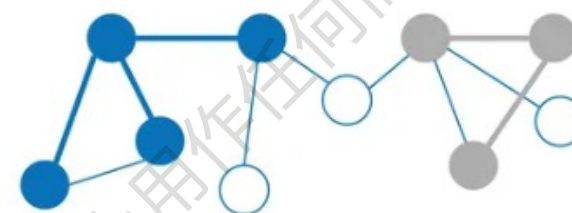
Pushes Labels to Neighbors to Find Clusters



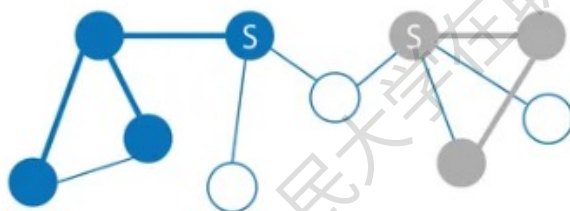
Example of 2 nodes given seed labels.



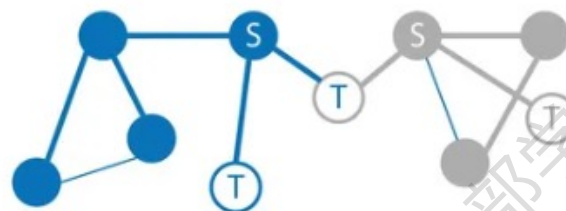
They look for immediate neighbors as targets to spread their label to.



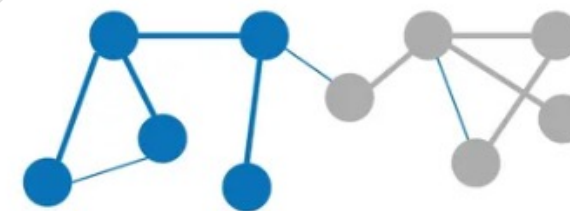
Where there is no conflict the label spreads.



The recently labeled nodes now activate like a new seed.



Conflicts are resolved based on a set measure such as relationship weights.



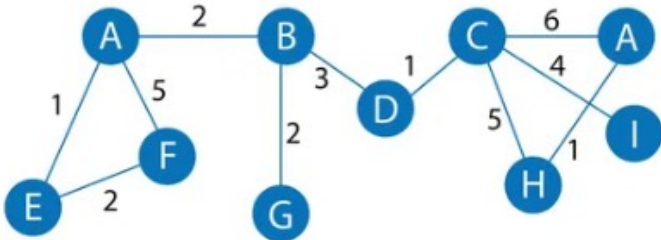
The process continues until all nodes are updated. 2 Clusters are identified.

LPA can be run with seed labels plus unlabeled nodes or each node starting with a unique label.
The more unique labels, the more conflict resolution used.

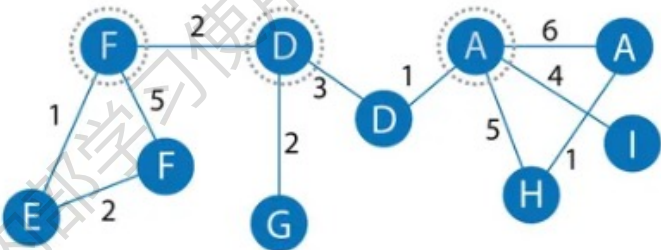
Source: <https://zhuanlan.zhihu.com/p/64984300>

Label Propagation - PULL

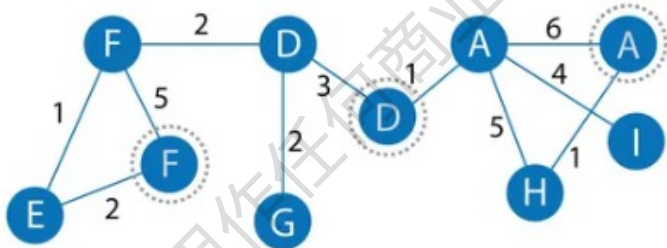
Pulls Labels from Neighbors based on Relationship Weights to Find Clusters



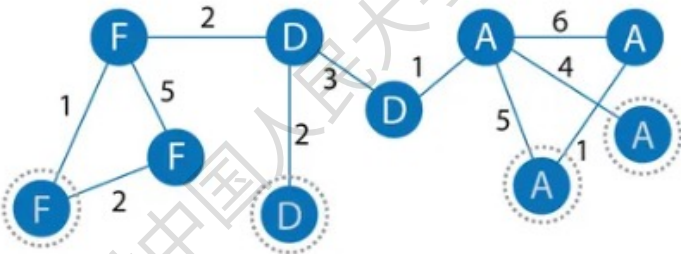
Example with 2 nodes with same label "A". All others are unique. Node weights default to 1 and in this example are ignored.



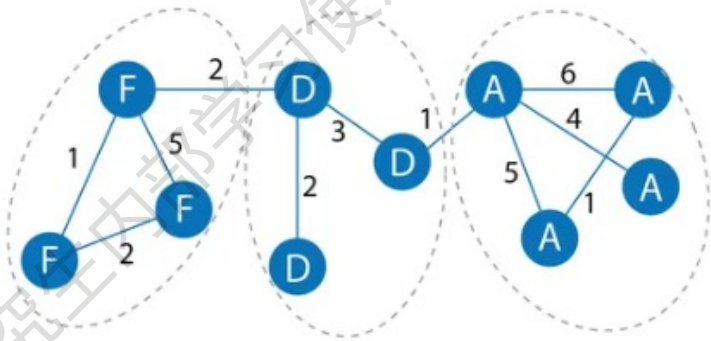
Nodes are shuffled for processing order and each node considers its direct neighbors' labels. (3 highlighted above.) Nodes acquire the label matching the total highest relationship weights.



Note that these 3 nodes don't change labels because their highest weight relationships in this step have the same label.



This continues until all nodes have updated their labels.



3 Clusters are identified. The labels themselves have no meaning. Source: <https://zhuanlan.zhihu.com/p/64984300>

- **模块度 (Modularity)** : Newman在2003年提出, 用来衡量一个社区的划分是否优良。

一个好的划分结果其表现形式是: 在社区内部的节点相似度较高, 而在社区外部节点的相似度较低。

Q就是模块度, 模块度越大则表明社区划分效果越好

$$Q = \sum_c \left[\frac{\sum_{in}}{2m} - \left(\frac{\sum_{tot}}{2m} \right)^2 \right]$$

其中, $\sum_{in}$ 表示的是社区 c 内部的权重, $\sum_{tot}$ 表示的是与社区 c 内部的点连接的边的权重, 包括社区内部的边以及社区外部的边。



早期的算法不能够很好的确认什么样的社区划分是最优的。Modularity这个概念就是为了定义一个对于社区划分结果优劣的评判。

- 模块度定义不同
→不同的社区发现算法

Ref.	Modularity	Network Type		Community Type
		Directed	Weighted	
(Newman 2004a)	$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(C_i, C_j)$	No	No	Disjoint
(Newman and Girvan 2004)	$Q = \sum_{k=1}^s \left[\frac{l_k}{L} - \left(\frac{d_k}{2L} \right)^2 \right]$	No	No	Disjoint
(Newman 2004a)	$Q = \frac{1}{2W} \sum_{i,j} \left[A_{ij} - \frac{s_i s_j}{2W} \right] \delta(C_i, C_j)$	No	Yes	Disjoint
(Fortunato 2010)	$Q = \sum_{c=1}^{n_c} \left[\frac{W_c}{W} - \left(\frac{S_c}{2W} \right)^2 \right]$	No	Yes	Disjoint
(Arenas et al. 2007)	$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i^{\text{out}} k_j^{\text{in}}}{m} \right] \delta(C_i, C_j)$	Yes	No	Disjoint
(Arenas et al. 2007)	$Q = \frac{1}{2W} \sum_{i,j} \left[W_{ij} - \frac{k_i^{\text{out}} k_j^{\text{in}}}{W} \right] \delta(C_i, C_j)$	Yes	Yes	Disjoint
(Shen et al. 2009)	$Q = \frac{1}{2m} \sum_{i,j} \frac{1}{o_i o_j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(C_i, C_j)$	No	No	Overlapping
(Nicosia et al. 2009)	$Q = \frac{1}{m} \sum_{c=1}^{n_c} \sum_{i,j} \left[r_{ijc} A_{ij} - s_{ijc} \frac{k_i^{\text{out}} k_j^{\text{in}}}{m} \right]$	Yes	No	Overlapping

- **Louvain 模块度算法**：该算法在效率和效果上都表现较好，并且能够发现层次性的社区结构，适合庞大网络的社群发现。

□ 算法思想

Louvain算法包括两个阶段：

- **步骤一**，不断地遍历网络中的结点，尝试将单个结点加入能够使modularity提升最大的社区中，直到所有结点都不再变化。
- **步骤二**，处理第一阶段的结果，将一个个小的社区归并为一个超节点来重新构造网络，这时边的权重为两个节点内所有原始节点的边权重之和。
- **迭代这两个步骤**直至算法稳定。

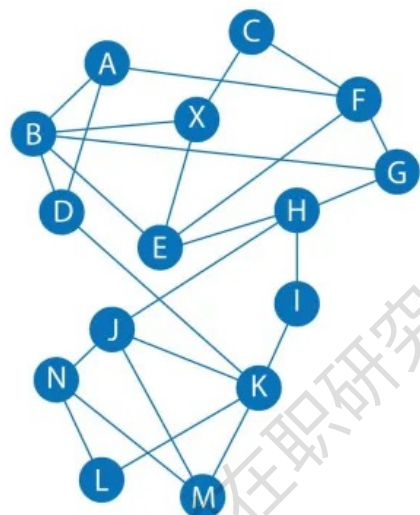


Louvain算法是一个迭代算法，每一轮迭代都会产生一个当前局部最优的社区结构。

Louvain Modularity Algorithm



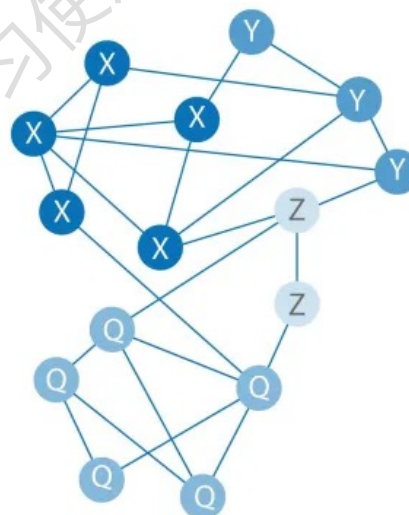
第一步迭代:



Step 0

Choose a start node (X above) and calculate the modularity options for joining different communities with each of its immediate neighbors.

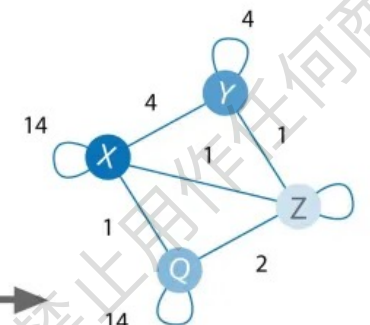
Pass 1



Step 1

The start node joins the node with the highest modularity change. the process is repeated for each node with the above communities formed.

超节点网络



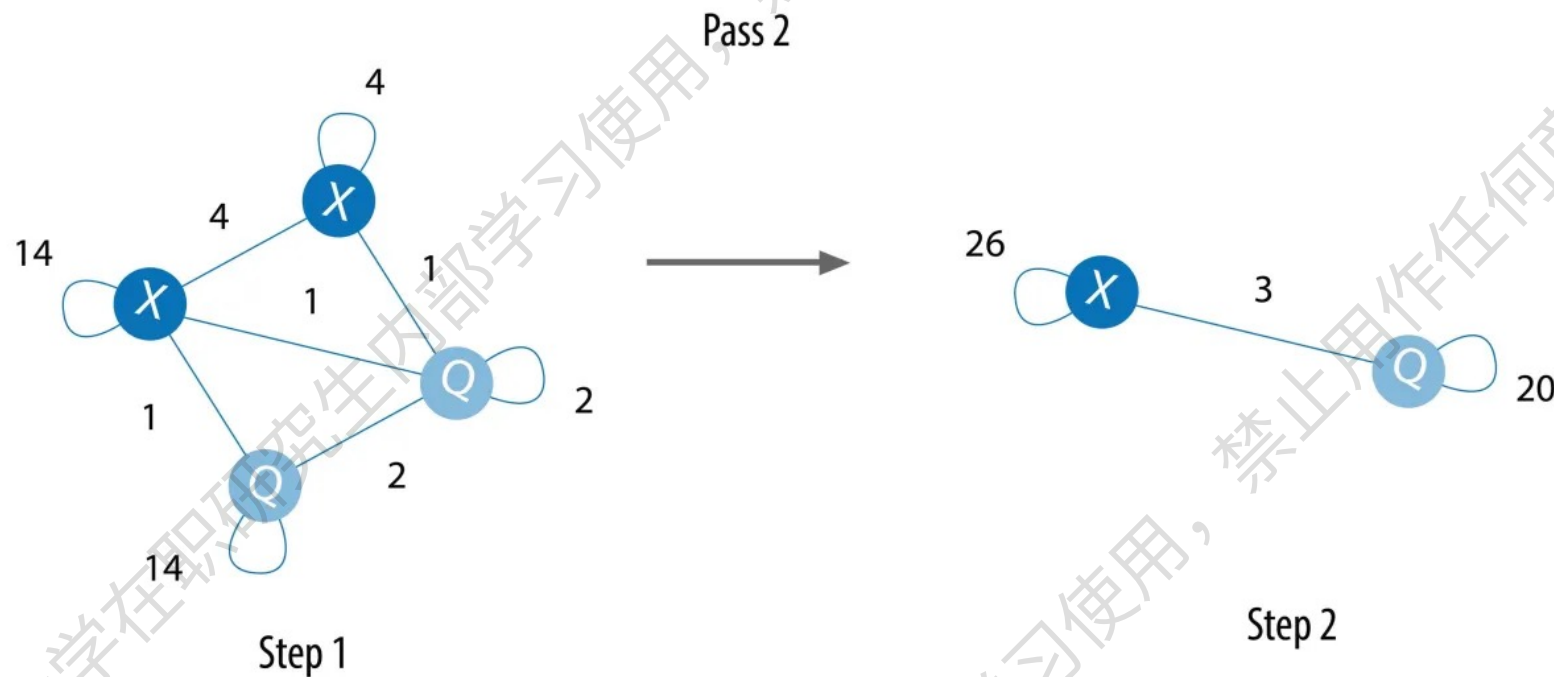
Step 2

Communities are aggregated to create super communities and the relationships between these super nodes are weighted as a sum of previous links. (Self-loops represent the previous relationships in both directions now hidden in the super node)

Louvain Modularity Algorithm



第二步迭代:



Steps 1 and 2 repeat in passes until there is not further increase in modularity or a set number of iterations have occurred.

- 在你关心的网络中，运行社群发现算法有什么意义？
- 说一个你了解的社群发现算法的应用



□ 金融风险

- 在金融风险场景中根据用户行为特征进行团伙识别。

□ 社交网络

- 基于网络关系中点对（一条边连接的两个顶点）之间关联的广度和强度进行社交网络划分；对复杂网络分析、电话网络分析人群之间的联系密切度。

□ 推荐系统

- 基于用户兴趣爱好的社区发现，可以根据社区并结合协同过滤等推荐算法进行更精确有效的个性化推荐。

A network-centric approach for price prediction



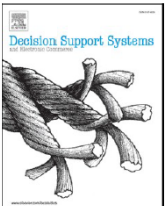
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Decision Support Systems xxx (xxxx) xxx

Contents lists available at [ScienceDirect](#)

Decision Support Systems

journal homepage: www.elsevier.com/locate/dss



LSTM-ReGAT: A network-centric approach for cryptocurrency price trend prediction

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ARTICLE INFO

Keywords:
Cryptocurrency
Price trend prediction
Cryptocurrency network
Graph attention network

ABSTRACT

Predicting price trends of cryptocurrencies is a challenging task due to the highly speculative market. Prior studies mainly investigate predictors such as historical trading data, macroeconomy and public interests in cryptocurrencies for price trend prediction while ignoring the predictability between cryptocurrencies. In fact, the price movement of a cryptocurrency may be affected by other cryptocurrencies, thus, incorporating cryptocurrency interrelations can further improve performance. In this paper, we propose a novel end-to-end model for price trend prediction using long short-term memory (LSTM) and relationwise graph attention network (ReGAT), in which both individual cryptocurrency features and cryptocurrency relations are considered. Specifically, a cryptocurrency network is built by using shared features between cryptocurrencies. Further, LSTM is used to profile sequential patterns of individual cryptocurrency features. To fully extract network features, ReGAT is designed to aggregate information conveyed by different types of relations while automatically differentiating their importance. Finally, the conventional and network features are concatenated to predict the price trend. The effectiveness of LSTM-ReGAT is validated using real-world cryptocurrency market data. The trading simulations for Bitcoin and portfolios reveal that our model obtains the highest profits. Our study provides insightful implications for investment decision support in cryptocurrency market.

Table 1
Features for cryptocurrency network construction.

Cryptocurrency interrelations	Features
Technological foundation	Algorithm: PoW, Hybrid-PoW&PoS, PoS, Script, SHA-256, etc. Platform: BNB Chain, Ethereum Ecosystem, Cosmos Ecosystem, BNB Smart Chain, Polygon Ecosystem, etc.
Industry	Industry: DeFi, Payments, Medium of Exchange, Enterprise Solutions, Web3, etc.
Investor co-attention	Investment portfolio: Kenetic Capital Portfolio, Pantera Capital Portfolio, DCG Portfolio, Alameda Research Portfolio, Hashkey Capital Portfolio, etc.

A network-centric approach for price prediction

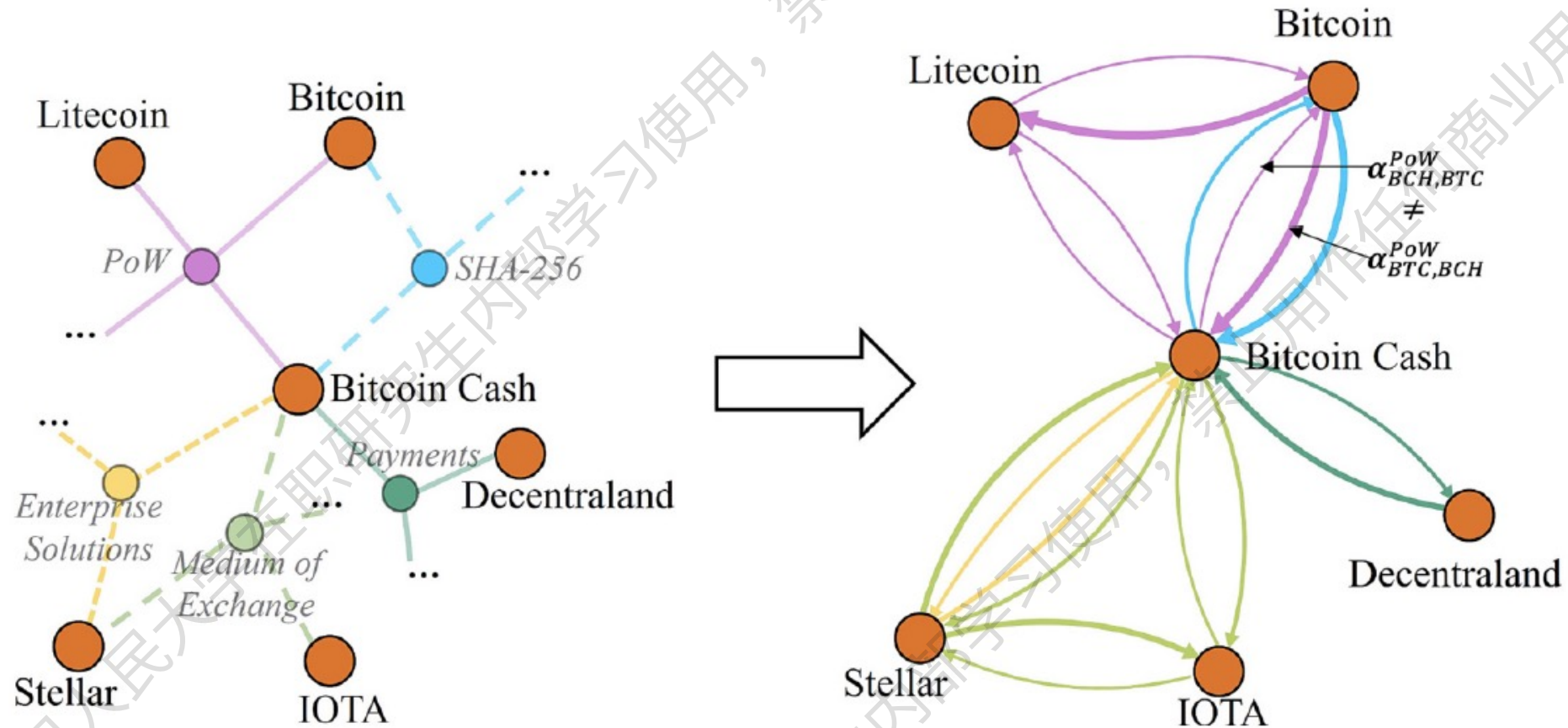


Fig. 2. An example cryptocurrency network.

- Adding diversification in cryptocurrencies helps increase the Sharpe ratio and reduce the total risk [29].
- In our cryptocurrency network, the closely connected nodes have strong risk correlations because they share the same attributes. Therefore, closely connected cryptocurrencies in the network should avoid being selected into the same portfolio.

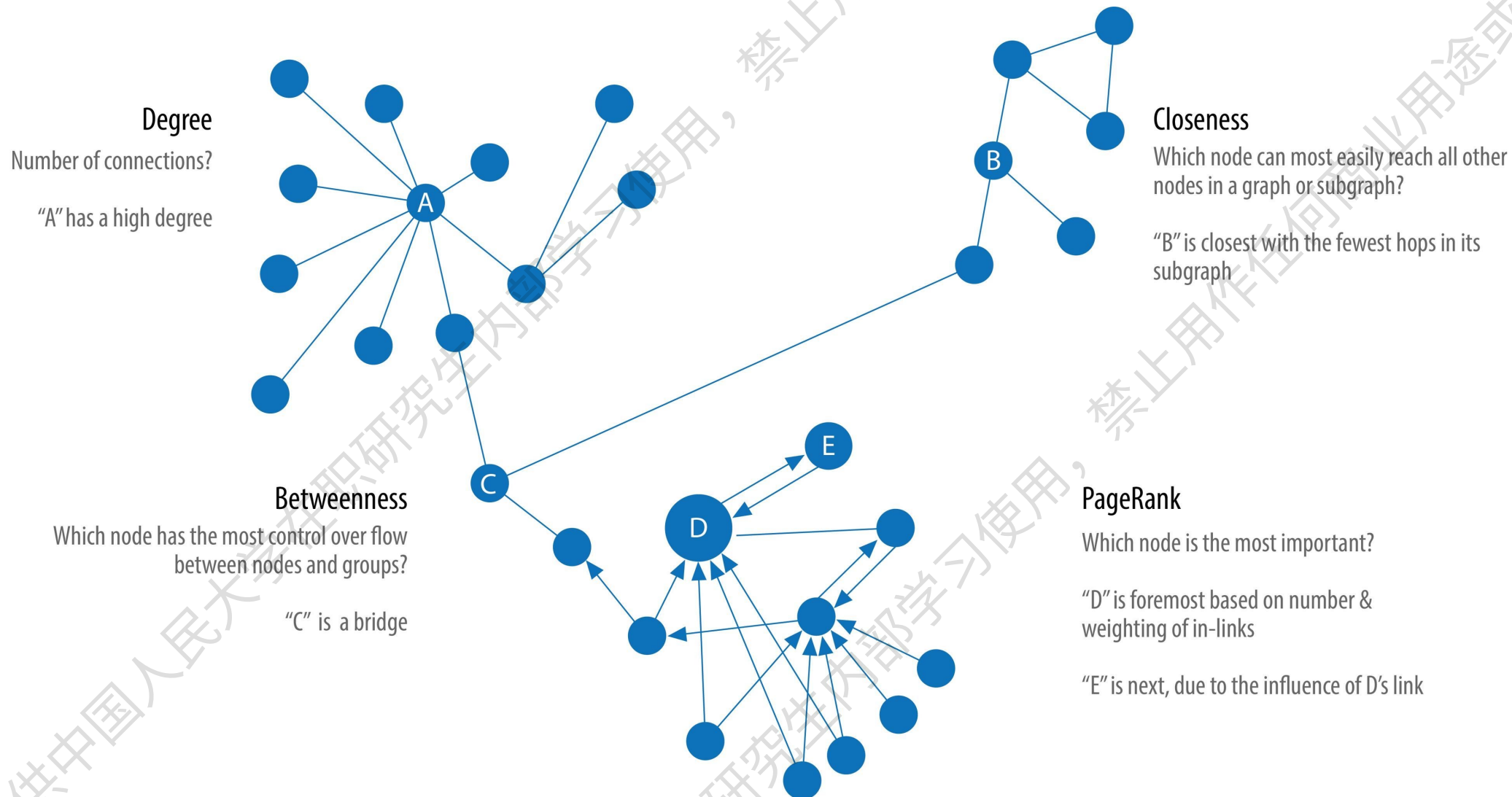
A popular community detection algorithm, Louvain [5], is used to identify closely connected cryptocurrencies. Louvain is an unsupervised community detection algorithm based on modularity optimization. We chose Louvain because it does not require specification of the number of communities or the expected size of community. Through Louvain, the network is divided into 7 communities, containing 252, 153, 131, 49, 33, 22, 5 respectively. Cryptocurrencies within the same community are treated as sharing similar risks. Therefore, we set the number of cryptocurrencies in a portfolio as $C=7$. Cryptocurrency c has a probability $\eta_{c,t}$ of being selected in the portfolio at day t . Total balance is allocated to each cryptocurrency in the portfolio by weight $w_{c,t}$. Cryptocurrency c is decided whether to be bought or not with a probability $p_{c,t}$. Here, we used three baseline strategies and the cryptocurrency network separately to create investment portfolios. The details are provided as follows.

Table 8

Trading simulation results of portfolios.

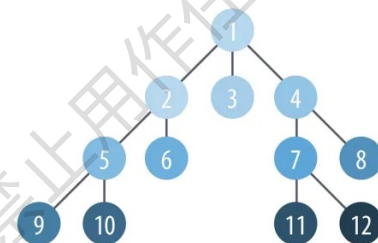
Trading strategy	Average final balance (Average volatility)	
	Shorting not allowed	Shorting allowed
Random Strategy-Portfolio	0.870 (0.0298)	0.975 (0.0388)
Momentum Strategy-Portfolio	0.794 (0.0278)	0.898 (0.0358)
Baseline Strategy-Portfolio	1.220 (0.0277)	1.382 (0.0349)
Cryptocurrency Network-Portfolio	1.307 (0.0255)	1.480 (0.0327)

基础算法总结-中心性

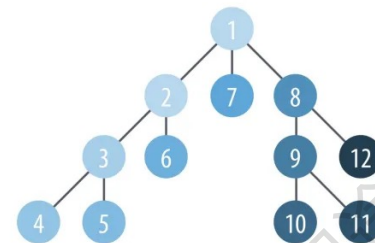


基础算法总结-路径搜索

- 图搜索
- 路径搜索
- Random walk

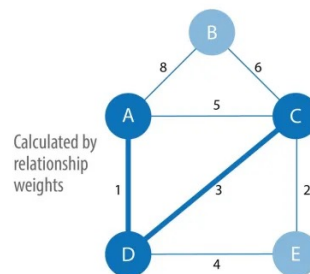


Breadth First Search
Visits nearest neighbors first



Depth First Search
Walks down each branch first

Pathfinding Algorithms



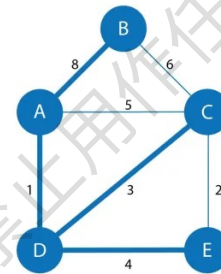
Shortest Path

Shortest path between 2 nodes (A to C shown)

(A, B) = 8
(A, C) = 4 via D
(A, D) = 1
(A, E) = 5 via D
(B, C) = 6
(B, D) = 9 via A or C
And so on...

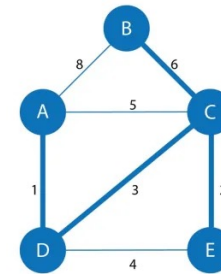
All-Pairs Shortest Paths

Optimized calculations for shortest paths from all nodes to all other nodes



Single Source Shortest Path

Shortest path from a root node (A shown) to all other nodes

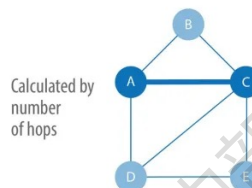


Minimum Spanning Tree

Shortest path connecting all nodes (A start shown)

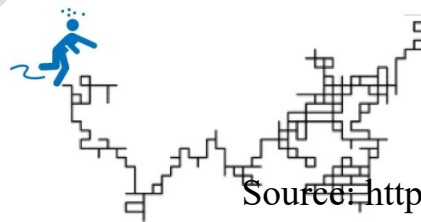
Traverses to the next unvisited node via the lowest cumulative weight from the root

Traverses to the next unvisited node via the lowest weight from any visited node



Calculated by number of hops

Random Pathfinding Algorithm



Random Walk

Provides a set of random, connected nodes by following any relationship, selected somewhat randomly.

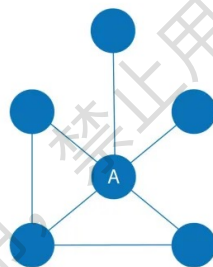
Also called the drunkard's walk.

Source: <https://zhuanlan.zhihu.com/p/64984300>

基础算法总结-路径搜索

- 聚类系数
- 关联类算法
 - 标签传播算法
 - Louvain模块度算法

Measuring Algorithms



Triangle Count

The number of triangles that pass through a node. A has 2 triangles.

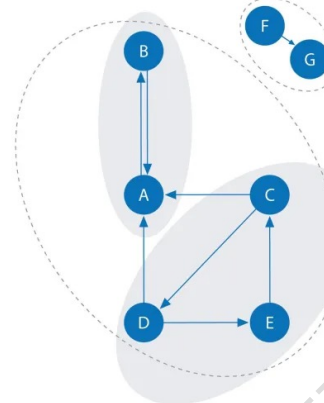
Clustering Coefficient

The probability that the neighbors of a node are connected to each other.

A has a 0.2 CC. Any 2 nodes connected to A have a 20% chance of being connected to each other.

These measures can be counted/normalized globally

Components Algorithms



Connected Components

Sets where all nodes can reach all other nodes, regardless of direction.

2 Sets shown with dashed outlines {A,B,C,D,E} and {F,G}

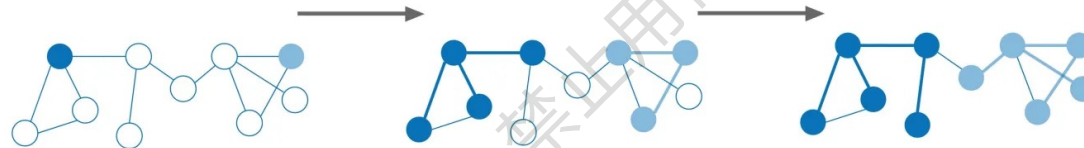
Strongly Connected Components

Sets where all nodes can reach all other nodes in both directions, but not necessarily directly.

2 Sets shown shaded {A,B,C} and {D,E}

Label Propagation Algorithm

Spread Labels To or From Neighbors to Find Clusters.



Run over multiple iterations.

Weights of relationships and/or nodes are often used to determine label "popularity" in a group.

Louvain Modularity Algorithm

Find clusters by moving nodes into higher relationship density groups and aggregating into super communities.

